Zhuk's Simplified CSP Dichotomy Proof

A source audit

Abstract

The companion proof paper corrects [14] silently: where the source is wrong, incomplete or ambiguous, it states the corrected thing and proves it, with nothing in the margin about what the source has. This document is the record of those departures. For each it says what the source has, quotes it, says why it cannot be transcribed, gives the repair, and names the statement of the proof paper that carries it.

Sixteen substantive defects, of which fifteen are repaired with no new mathematics and one — the maximal multi-type extension — is reduced to a single stated hypothesis that remains unproved. Ten readings legislated, two of which turn out to be theorems rather than choices. Two claims raised and refuted, one declared moot. Four hedges in §5 of the source, all four honest.

References of the form `lem:...` are labels of statements in the proof paper, which is compiled separately; the label is the join key between the two documents.

Contents

1 What this document is

The proof paper is written in the Jordan–Schönflies mould: where [14] is wrong, incomplete, or ambiguous, the proof paper simply states the corrected thing and proves it, with no marginal notes about what the source has. That is the right shape for a document whose job is to be checked as mathematics. It is the wrong shape for a reader who wants to know *where our rendering differs from Zhuk's paper and why*, and it would be dishonest to make those differences invisible.

This document is the difference. For every place where the proof paper departs from [14], there is an entry here that says what the source has, quotes it, says why it cannot be transcribed, gives the repair, and names the statement of the proof paper that carries it.

None of this is a criticism of [14], which is unusually careful prose — across nine thousand lines of TEX it says "obvious" three times and "clearly" never, and a mechanical sweep of §5 turns up four hedges in thirteen printed pages, all four of which turn out to be honest (§5.2). Sixteen substantive defects in fifty pages is roughly the rate one should expect of any informal exposition of a proof this size, and finding them is the point of reading it this way.

1.1 The evidentiary standard

A claim counts as checked here only when the cited lines *and the prose around them* have been read, and the passage that does or does not discharge the claim can be quoted. The standard is not pedantry: two of the original claims died to it. One was taken from a reader who worked from a pseudocode block's `Input:` line while the hypothesis it needed was in the paragraph above (§6.2); the other asserted a citation cycle that a parser refuted in minutes (§6.1). Both failure modes are the same: reading the formal-looking part of a paper and not the sentence before it.

Where a claim is reducible to a finite check, it has had one. The searches are listed in §7 with their scope, so that a reader can see what they do and do not cover.

1.2 Line numbers, and how entries are keyed

Line numbers are into `main.tex` and `StrongSubalgebras.tex` of the v2 arXiv source of [14], obtainable from `arxiv.org/e-print/2404.01080v2`. Zhuk's paper is not redistributed here, and its statement names are its own: `LEMPropagation`, `THMMainStableIntersection` and so on are the source's labels, not ours.

Each entry names the statement of the proof paper that carries the repair, by that statement's label, set in typewriter: `lem:propagation`, `thm:stable-intersection`. The two documents are compiled separately and share no cross-reference machinery, deliberately, so that neither can be read as evidence for the other. The label is the join key.

The tags D2, D4, . . . are historical: they were assigned in the order the claims were raised, and some early tags were retired when the claims they named were refuted or restated. The register of §2 is the current and complete list; the retired claims are in §6.

1.3 Where things stand

	Count
Defects confirmed and repaired, needing no new mathematics	15
of which: a citation the later paper dropped	3
a typographical or typing slip	2
a statement that is false as printed	2
omitted steps and dropped hypotheses	8
Defects confirmed and **open**	1
Claims examined and refuted	2
Claims examined and declared moot	1
Readings legislated (of which two are theorems, not choices)	10
Hedges in §5 of the source, all discharged	4

The single open item is D17 (§2.11), the maximal multi-type extension. It was declared solved in an earlier draft; an external review found the repair invalid at its first line, and the finding is correct. It is recorded here as open rather than quietly weakened.

Its status has since been sharpened rather than resolved. The proof paper now proves the lemma from an *explicitly stated* extra hypothesis — that the type-T factors alone already cut B_2 out — and shows that among those factors the image-versus-intersection failure does not occur, so the whole difficulty is now localised in one displayed sentence rather than spread through a proof. The hypothesis itself is unproved and is a debt at the single call site.

2 The register

Sixteen places where [14] is wrong, incomplete, or states less than its own proof gives. Fifteen are repaired and none of those repairs needs new mathematics; the sixteenth, D17, is open.

Tag	Where	What	Proof paper
D1	§2.3	Lemma 19 restates a corollary of [15] dropping two hypotheses; false as printed	`lem:ba-center-implies`
D2	`main:1682`	Propagation modulo a congruence: stated, used five times, never proved	`lem:propagation`
D4	`SS:2594`	$n = 2$ for type C in stable intersection is argued only for PC	`thm:stable-intersection`
D6	§3.2	Bridges-from-relations applied without its third hypothesis, which can fail	`lem:connected`
D7	§3.3	Two gaps in the main induction: *connected* proved where *linked* is demanded, and (1c) derived for the wrong reduction	`thm:main-inductive`
D8	`main:3526`	Minimal-containing-a-point is used where minimal-outright is required; the bridging lemma was deleted	`lem:minimal-consistent`
D11	`SS:1154`	The symmetrisation formula defines a relation that is never a bridge	`lem:symmetrise`
D12	`SS:1103`	Bridges and abelian groups: *false as stated*, with a three-element counterexample	`lem:nice-bridge-abelian`
D13	`SS:1273`	Linkedness asserted from an inclusion that bears on a different hypothesis	`lem:nontrivial-bridge`
D14	`SS:1394`	The box form of absorption-preserves-linkedness cited for a restriction that is not a box	`lem:absorb-linked-sub`
D15	`SS:2208`	Existence of an irreducible congruence factors by the wrong congruence, hiding an unsupported properness step	`lem:main-existence`
D16	`SS:2743`	Linear congruences on top: the conclusion is asserted	`lem:linear-on-top`
D17	`SS:3113`	Maximal multi-type extension concludes that two sets meet because each meets a third. **Open**	`lem:maximal-mult`
C10	`SS:208`	The central-relation citation drops the third alternative of the theorem it cites	`lem:central-relation`
F	`SS:545, 837, 921`	A full fibre inferred from a surjective projection, at three sites	`lem:full-fibre`
D18	`main:1373`	The definition of a linear congruence is ill-typed: it intersects a relation on elements with a set of tuples of classes	`def:linear-congruence`

Two patterns recur rather than occurring once, and are worth naming because a reader checking the proof paper will meet them repeatedly. The first is a *proper* containment asserted where the cited lemma gives only $\subseteq$, usually after factoring by a congruence, which can close the gap; it appears at least five times and is repaired the same way each time, by producing the witness the source has already displayed two lines earlier. The second is an upgrade from "the image has size one" to "the image is *this* block"; it appears four times and the proof paper makes it a named step.

2.1 D1 — two hypotheses dropped, and the statement becomes false

[14, Lem. 19] restates [15, Cor. 6.1.2, 6.9.2]. The source it cites reads "Suppose $R \leq A_1 \times \cdots \times A_n$, $\mathrm{pr}_1(R) = A_1$, $B_i \leq_{\mathrm{BA}(t)} A_i$ for every i". The restatement drops the subdirectness

$\mathrm{pr}_1(R) = A_1$, and drops — in the BA case — that the B_i absorb with a *common* term t, which is the reason [15] carries t in the notation.

Without the first the statement is false. Take $R = \{(0,0)\} \leq \mathbf{Z}_p \times \mathbf{Z}_p$, a subuniverse by idempotence, and $C_i = A_i$, which is allowed because $\leq_T$ admits equality. Then $R \cap (C_1 \times C_2) = R$ and $\mathrm{pr}_1(R) = \{0\}$, which is neither empty nor all of $\mathbf{Z}_p$; the conclusion therefore asserts $\{0\} <_{\mathrm{BA}} \mathbf{Z}_p$, contradicting the paper's own Lemma 29.

Repair. Restore both hypotheses. The proof paper states them in `lem:ba-center-implies`. The common-term condition is not free at the call site: `thm:main-inductive`(2) records only that every coordinate reduction has type BA, and applies the lemma to the relation computing E_2 from E_1. Either a lemma synthesising one common binary term from a finite family of coordinate absorptions is needed, or the reduction data must carry a common witness from the start. This is recorded as an obligation on the proof paper, not as a defect of the source, since the source's own version of the lemma carries the hypothesis.

2.2 D2 — propagation modulo a congruence is never proved

Stated at `main:1682`; used at `SS:2506, 2514, 2588, 3110` and `main:2999`. Never proved, and not attributed to any external source.

The paper's own convention (`main:1058`) is *"we always duplicate statements if the proof appears in a later section"*, implemented by `\newtheorem*` duplicates. Checking all thirteen statements of §2.3 against that convention leaves exactly two not restated in §5: this corollary, and `LEMBACenterImplies`, which is declared an external import and so is absent by design.

Repair. It is Propagation specialised to the canonical surjection $\pi \colon \mathbf{A} \twoheadrightarrow \mathbf{A}/\delta$, whose clauses (f), (ft), (fs), (fm) are its four clauses. The identification is literal rather than approximate: X/δ is *defined* as the image $\{x/\delta : x \in X\}$, which is $\pi(X)$; in particular the corollary's hypothesis "B/δ is S-free" and Propagation's "$f(B)$ is S-free" are the same condition. Proof paper: `cor:prop-quotient`, derived from `lem:propagation`.

A mechanical audit of all 81 labelled statements of the source (§7) shows this is the *only* statement that is cited while being neither proved nor attributed. That matters beyond this entry: the audit covers §5.6 and §5.7, so whatever else is wrong there, it is not another statement asserted from nowhere.

2.3 D4 — $n = 2$ for type C is unargued

The stable-intersection theorem concludes, in its case (c), that $n = 2$ and $T_1 = T_2 = \mathrm{C}$. The proof at `SS:2594` ff. reduces general n to a pair, obtains $T_1 = T_2$, and then:

> "If $T_1 = \mathrm{PC}$ then $\sigma_1 \circ \sigma_2 = \sigma_1 = \sigma_2 \ldots$ Additionally, this implies that n cannot be greater than 2 for $T_1 = \mathrm{PC}$, as in this case the intersection $C_1 \cap C_2$ must be empty."

That argument is specific to PC: equal congruences make C_1 and C_2 distinct blocks, hence *pairwise* disjoint, which contradicts hypothesis (3) once $n \geq 3$. For type C there is no congruence and no analogue, and the proof offers nothing; hypothesis (2) gives only $C_1 \cap \cdots \cap C_n = \varnothing$, never $C_1 \cap C_2 = \varnothing$. The same gap is visible at `SS:2650`, where the PC argument needs one further unstated step — the two σ-blocks are distinct because $B_1 \cap C_2' \neq \varnothing$ would otherwise put a point in $C_1 \cap C_2'$ — which again has no type-C analogue.

Repair. The missing argument is [15, Thm. 3.7], which is the same statement for subuniverses of a *single* algebra, resting on [15, Lem. 6.11]: for $n \geq 3$ and C_i central in A_i, no $(C_1, \ldots, C_n)$-essential relation $R \leq A_1 \times \cdots \times A_n$ exists. The 2024 setting is not literally that of Theorem 3.7 — there the B_i are subuniverses of one algebra, whereas here C_i is central in B_i and the B_i differ — so a reduction is needed first: put $D = \bigcap_i B_i$ and $E_i = C_i \cap D$; then D is a nonempty subuniverse, each E_i is properly central in D, $\bigcap_i E_i = \varnothing$

and $\bigcap_{i\neq j} E_i \neq \varnothing$ for every j. The diagonal of D^n is then $(E_1, \ldots, E_n)$-essential. Proof paper: inside `thm:stable-intersection`.

Cost of the repair. [15, Lem. 6.11] has a printed proof that cites *itself* three times where it means the preceding lemma, so a self-contained rendering must supply both. Both citations were re-read verbatim (Lemma 6.11 at 2005.00593 p. 37, Theorem 3.7 at p. 46).

2.4 D6 — a hypothesis that can fail, and cruciality supplies it

`LEMBridgeFromRelation` requires, besides subdirectness and rectangularity of the first and last variables, that there exist $(b_1, \bar{a}, a_n)$ and $(a_1, \bar{a}, b_n)$ in R with $(a_1, \bar{a}, a_n) \notin R$. The proof of `LEMConnectedProperties`(a) invokes it for every constraint along a path and never discharges that hypothesis.

It can fail. Machine-checked witness: $R = \{(x, z, y) \in \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_4 : x \equiv z \equiv y \pmod 2\}$ is subdirect, $\mathrm{Con}_1(R, 1)$ is congruence mod 2 and coordinate 1 is rectangular, and the third hypothesis is unsatisfiable. The construction then yields $\delta(x_1, x_2, y_1, y_2) = \exists z\, R(x_1, z, y_1) \wedge R(x_2, z, y_2)$, whose projection onto the first two coordinates *equals* $\mathrm{Con}_1(R, 1)$ rather than properly containing it — so δ is not a bridge, clause (c) failing. The lemma is genuinely inapplicable and genuinely necessary.

Repair. Cruciality supplies the hypothesis. If the third hypothesis fails at coordinates 1 and n then R is the conjunction of its two projections,

$$R = \{(x, \bar{a}, y) : (x, \bar{a}) \in \operatorname*{pr}_{[n-1]}(R) \text{ and } (\bar{a}, y) \in \operatorname*{pr}_{[n]\setminus\{1\}}(R)\},$$

the inclusion $\supseteq$ being exactly the failure of the hypothesis. A constraint that is crucial in a reduction therefore satisfies the hypothesis at every pair of its coordinates: otherwise weakening it replaces it by constraints whose conjunction defines the same relation, so the weakened instance has the same solution set and in particular still has no solution in the reduction, contradicting cruciality. On the witness above, $\mathrm{pr}_{1,2}(R)$ and $\mathrm{pr}_{2,3}(R)$ do conjoin to R — exactly the degenerate shape the corollary rules out.

Proof paper: `lem:connected` is stated for crucial instances. This is a notion [14] inherits from `LEMCrucialMeansIrreducible` and then stops using.

2.5 D7 — two gaps in the main induction

(i) The endgame proves *connected*; (1c) demands *linked*. Both endgames of Case 2 finish with "... which means that $\mathcal{I}$ is connected and satisfies (1b) if its solution set is subdirect or (1c) otherwise". But (1c) requires a *linked* connected subinstance, and only connectedness was established.

Repair, via irreducibility. Suppose $\mathcal{I}$ is connected with non-subdirect solution set. Take $\mathcal{J} = \Upsilon = \mathcal{I}$, legitimate since an instance is a weakening, hence an expanded covering, of itself. It remains to see $\mathcal{I}$ is linked. If it were not, $\mathcal{I}$ would witness the failure of its own irreducibility: its variables are among its own, each constraint is the projection of itself onto all its variables, it is not fragmented (connectedness of the constraint-adjacency graph forces constraints to share variables), it is not linked, and its solution set is not subdirect — precisely conditions (i)–(v) of the definition. Since $\mathcal{I}$ is irreducible by standing hypothesis, it is linked. Two lines, and they explain why irreducibility is a hypothesis of the theorem at all.

(ii) Case 1 derives (1c) for the wrong reduction. Case 1 obtains a 1-consistent $D^{(2)} \leq_T D^{(1)}$, shows $\mathcal{I}$ is crucial in $D^{(2)}$, and closes with "applying the inductive assumption to $D^{(2)}$ we derive the required conditions". Conditions (1a) and (1b) mention no reduction and transfer verbatim. (1c) does not: it asserts an expanded covering *crucial in the ambient reduction.*

The inductive hypothesis delivers $\mathcal{K}$ crucial in $D^{(2)}$; the goal needs a covering crucial in $D^{(1)}$, and the containment runs the wrong way — an instance with no solution in the smaller $D^{(2)}$ may have one in $D^{(1)}$.

First, the induction structure, which is load-bearing and unstated. The proof opens "We prove the claim by induction on the size of $D^{(1)}$. *Let us prove (2) first.*" That ordering is not stylistic. Part (2) at a given measure uses (1) only at strictly smaller measure, whereas part (1) at that measure uses **(2) at the same measure** — Case 1 does exactly this at `main:3507`. So the induction is: strong induction on $k = \mu(D^{(1)})$; at each level establish (2) for *all* instances, then (1) for all instances. That is acyclic: $(2)_k \leftarrow (1)_{<k}$, $(1)_k \leftarrow (2)_k, (1)_{<k}$.

Why that alone does not close the gap. One would like to apply (2) to $\mathcal{K}$ and conclude contrapositively that $\mathcal{K}^{(1)}$ has no solution. But $\mathcal{K}$ is an expanded *covering*, so it has more variables than $\mathcal{I}$, and under $\mu = \sum_x |D^{(1)}_x|$ its measure is *larger*, not equal. The device that works for a weakening does not work for a covering.

The repair, which avoids re-entering the induction. Put $\mathcal{L} = \mathcal{K} \wedge \mathcal{I}$. It is an expanded covering of $\mathcal{I}$, since $\mathcal{K}$ is one, $\mathcal{I}$ is a weakening of itself hence one, and the union of two expanded coverings is one. $\mathcal{L}^{(1)}$ has no solution, since $\mathcal{L}$ contains every constraint of $\mathcal{I}$ and $\mathcal{I}$ is crucial in $D^{(1)}$. So $\mathcal{L}$ may be weakened to an instance $\mathcal{L}'$ crucial in $D^{(1)}$. Every constraint of $\mathcal{K}$ survives that weakening: each is crucial in $D^{(2)}$, so weakening it yields a solution in $D^{(2)} \subseteq D^{(1)}$, and weakening it inside any weaker instance yields only more solutions; so each is already crucial in $D^{(1)}$ at every stage, and the process only weakens constraints that are not. Therefore $\Upsilon \subseteq \mathcal{K} \subseteq \mathcal{L}'$, its properties being intrinsic to Υ and untouched, and $\mathcal{L}'$ is an expanded covering of $\mathcal{I}$ crucial in $D^{(1)}$. No appeal to the induction at a larger measure, so the measure survives intact.

Proof paper: `thm:main-inductive` and `haz:case1c`.

Entry 2.1 (A caveat on the measure)**.** The measure $\mu(D) = \sum_x |D_x|$ is well founded for the appeals the repair above uses, all of which are at a strictly smaller reduction on the *same* variable set. It is *not* well founded across expanded coverings, where the variable set grows and μ grows with it. The proof paper does not yet resolve that, and it is not a defect of the source alone; it is recorded here so that the two questions are not confused. A repair must make the measure insensitive to duplicated covering variables — either by formulating the theorem for an instance equipped with a map to a fixed base instance and measuring on the base variables, or by carrying a parent-domain profile and ordering those. Putting the quantifier over instances inside the induction, which is necessary for other reasons, does not by itself solve it.

2.6 D8 — minimal-containing-a-point versus minimal-outright

At `main:3526` the main induction picks, for each variable, "the minimal $D^{(2)}_x \leq_{\mathcal{M}T} D^{(1)}$ containing $s(x)$", and two lines later invokes a lemma whose hypothesis (5) requires $D^{(2)}_x$ to be *a minimal $\mathcal{M}T$ subuniverse* — minimal outright. Those are not the same condition. The lemma bridging them has been deleted, along with its citation:

```
3526  For every variable x choose the minimal D^(2)_x <=_{MT} D^(1) containing
s(x).
3527  %By Lemma \ref{LEMMinimalContainingIsMinimal}
3528  %$D^{(2)}_{x}$ is a minimal \mathcal{M}T subuniverse.
3529  By Lemma \ref{LEMMultiTypeStillStable}
```

The deleted statement (`main:1864`) was: if C is inclusion-minimal with $b \in C$ and $C \leq_{\mathcal{M}T} B$, then no $C' \subsetneq C$ has $C' \leq_{\mathcal{M}T} B$.

Repair. Fix $T \in \{\text{L}, \text{PC}, \text{D}\}$ and $B \lhd\lhd\lhd A$, let Σ be the finite set of congruences that are T-dividing for B, and for $b \in B$ put $M(b) = B \cap \bigcap_{\sigma \in \Sigma} [b]_\sigma$, discarding any σ with $B \cap [b]_\sigma = B$.

Then $M(b)$ is the least $\mathcal{MT}$ subuniverse of B containing b; and if $c \in M(b)$ then $[c]_\sigma = [b]_\sigma$ for every $\sigma \in \Sigma$, so $M(c) = M(b)$. Consequently an inclusion-minimal $\mathcal{MT}$ set containing b equals $M(b)$ and admits no proper $\mathcal{MT}$ subset: any such C' is nonempty, and any $c \in C'$ has $M(c) = M(b) \subseteq C'$.

The content is the second sentence: $b \mapsto M(b)$ is a closure whose image consists of pairwise disjoint minimal sets, because membership in $M(b)$ is decided blockwise and blocks of an equivalence relation either coincide or are disjoint. Proof paper: `lem:minimal-consistent` and `haz:minimality`.

2.7 D11–D13 — three defects, one missing hypothesis

§5.4 of [14] is the only part of §5 that manipulates bridges as objects in their own right rather than consuming them, and all three of its defects are about which auxiliary property survives a construction. All three are cured by one hypothesis, which the source already has in its vocabulary: `LEMPCCongruencePropertyInductiveStep` (`SS:1379`) carries, as hypothesis (2), exactly *pair-reflexivity* — $(a, b, a, b), (b, a, b, a) \in \delta$ for every $(a, b) \in \mathrm{pr}_{1,2}(\delta)$.

D11: the symmetrisation formula. The proof of `LEMBlockOfGoodBridgeDoesNotHaveBAC` opens by symmetrising δ:

```
1154  \sigma(x_1,x_2,x_3,x_4) = \exists x_5 \exists x_6
1155  \delta(x_1,x_2,x_5,x_5)
1157  \wedge \delta(x_3,x_4,x_5,x_6).
```

Clause (d) of the bridge definition (`main:1258`) is "$(a_1, a_2, a_3, a_4) \in \delta$ implies $(a_1, a_2) \in \sigma_1 \iff (a_3, a_4) \in \sigma_2$". The first conjunct has x_5 in both of the last two places, so $(x_5, x_5) \in 0_A$ forces $x_1 = x_2$. Hence $\mathrm{pr}_{1,2}(\sigma) \subseteq 0_A$, while clause (c) demands $\mathrm{pr}_{1,2}(\sigma) \supsetneq \sigma_1 = 0_A$: *the relation defined is never a bridge*, for any σ_1, and every later line of the proof is false of it.

It is a one-character slip. The intended relation is $\delta * \delta^{-1}$, and the source writes exactly that, correctly, 210 lines later at `SS:1365-1367`, for the same purpose.

D12: bridges and abelian groups, false as stated. `LEMNiceBridgeGivesAbelianGroup` (`SS:1103`): for σ a congruence and δ a bridge from σ to σ with $\mathrm{pr}_{1,2}(\delta) = \widetilde{\delta} = A^2$ and δ symmetric, there is an abelian group G with $(A/\sigma; \delta/\sigma) \cong (G; x_1 - x_2 = x_3 - x_4)$.

Counterexample. $\mathbf{A} = \mathbf{Z}_3$ with $w(x_1, \ldots, x_4) = x_1 + x_2 + x_3 + x_4 \bmod 3$ — idempotent since $4 \equiv 1$, and a special WNU, so $\mathbf{A} \in \mathcal{V}_4$. Take $\sigma = 0_{\mathbf{A}}$ and $\delta = \{(x_1, x_2, x_3, x_4) \in \mathbb{Z}_3^4 : x_1 - x_2 + x_3 - x_4 = 0\}$. Verified by exhaustion: δ is a subuniverse of A^4; clause (d) holds; $\mathrm{pr}_{1,2}(\delta) = \widetilde{\delta} = A^2$; δ is symmetric; and none of the six bijections $\mathbb{Z}_3 \to \mathbb{Z}_3$ carries δ to $\{x_1 - x_2 = x_3 - x_4\}$. Since every abelian group of order 3 is $\mathbb{Z}_3$, no G works.

What is true. δ contains all (a, a, b, b), so it is the preimage of some $K \leq G \times G$ under $(x_1, x_2, x_3, x_4) \mapsto (x_1 - x_2, x_3 - x_4)$. Clause (d) makes K meet $G \times 0$ and $0 \times G$ only at the origin, and subdirectness makes it a graph, so $\delta = \{\varphi(x_1 - x_2) = x_3 - x_4\}$ for an automorphism φ. Symmetry says $\varphi^2 = \mathrm{id}$; Zhuk's conclusion is the case $\varphi = \mathrm{id}$, and the counterexample is $\varphi = -\mathrm{id}$, which is why it needs p odd. The step that breaks is "composing the bridge δ with itself we get a bridge $\delta_0 \supseteq \delta$": $\delta * \delta = \{\varphi^2(x_1 - x_2) = x_3 - x_4\}$, which for $\varphi = -\mathrm{id}$ does not contain δ, and the closing count $|\delta| = |\delta_0| = |A|^3$ needs the containment it does not have.

D13: linkedness is asserted, not shown. `SS:1273` puts $\delta' = \delta \cap B^4 \cap (\sigma^* \times \sigma^*)$ for B a block of LeftLinked(σ^*) and says "Since $\widetilde{\delta} \supseteq \sigma^*$, δ' satisfies all the conditions of Lemma `LEMBlockOfGoodBridgeDoesNotHaveBAC`". Two of those conditions are that $\mathrm{pr}_{1,2}(\delta')$ properly contains 0_B and is *linked*. What $\widetilde{\delta} \supseteq \sigma^*$ gives is the third condition. Nothing offered bears on linkedness — and it does not follow: the witness (x_3, x_4) supplied by minimality of σ^* lies in

σ^* but in no particular block, and if it lands outside B for every choice then $\mathrm{pr}_{1,2}(\delta')$ collapses to 0_B and δ' is not even a bridge. The same gap recurs at `SS:1290`.

The single cure, and why it costs nothing. Pair-reflexivity kills all three. D13: $\sigma^* \subseteq \mathrm{pr}_{1,2}(\delta)$ by minimality of σ^*, since $\mathrm{pr}_{1,2}(\delta)$ is a subalgebra of A^2, stable under σ by clause (a) and properly containing σ by clause (c); so pair-reflexivity puts (x_1, x_2, x_1, x_2) in δ' for every $(x_1, x_2) \in \sigma^* \cap B^2$, giving $\mathrm{pr}_{1,2}(\delta') = \sigma^* \cap B^2$ exactly — which is linked precisely because B is a block. D12: pair-reflexivity *is* $\varphi = \mathrm{id}$, and it also repairs the proof directly, since $\delta \subseteq \delta * \delta$ by taking (x_1, x_2) itself as the intermediate pair. D11: $\delta * \delta^{-1}$ is pair-reflexive by construction, so the corrected formula delivers what the lemma needs.

And it costs nothing, because the two lemmas that need the hypothesis are each invoked exactly once in the whole paper — `LEMNontrivialReflexiveBridgeImplies` at `SS:1368`, on a $\delta * \delta^{-1}$, and `LEMNiceBridgeGivesAbelianGroup` from inside that call.

Proof paper: `def:pair-reflexive`, `lem:symmetrise`, `lem:nice-bridge-abelian`, `lem:nontrivial-bridge`, `rem:phi-twist`.

2.8 D14 — the box form cannot reach the subrelation

`SS:1394`, inside Case 1 of `LEMPCCongruencePropertyInductiveStep`: with $B <_T A$, $|B| > 1$ and $\delta' = \delta \cap B^4$,

> "By Lemma `LEMBACenterLinkedness` $\mathrm{RightLinked}(\mathrm{pr}_{1,2,3}(\delta')) = B^2$."

`LEMBACenterLinkedness` ([3, Prop. 2.15(i)]) restricts a binary relation by a *box* $B_1 \times B_2$. Applied with $W = \mathrm{pr}_{1,2,3}(\delta)$, $B_1 = P \cap B^2$, $B_2 = B$ it yields linkedness of $\mathrm{pr}_{1,2,3}(\delta) \cap B^3$, which contains $\mathrm{pr}_{1,2,3}(\delta \cap B^4)$ and can be strictly larger: a tuple of δ whose first three coordinates lie in B need not have its fourth there. The two really do differ — for the meet-semilattice on $\{0, 1, 2\}$ with $B = \{0, 1\}$, the subalgebra $K = \{(0, 0), (1, 2)\}$ of A^2 has $\mathrm{pr}_1(K) \cap B = \{0, 1\}$ but $\mathrm{pr}_1(K \cap B^2) = \{0\}$. Getting $\delta \cap B^4$ out of the box form instead would need linkedness of δ in the split $(1, 2) \mid (3, 4)$ or $(1, 2, 3) \mid (4)$, and hypothesis (3) gives neither: a path in the coarser bipartite graph does not lift, because consecutive edges through a third coordinate may use different fourth coordinates. The lemma's own hypothesis $R \cap (B_1 \times B_2) \leq_{\mathrm{sd}} B_1 \times B_2$ is also never discharged.

The assertion is true; the citation is to the wrong form of the principle. What closes it is the *subrelation* form: if $W \leq_{\mathrm{sd}} \mathbf{P} \times \mathbf{A}$ is linked and $\varnothing \neq W' \leq_T W$ with $T \in \{\textsc{ba}, \textsc{c}\}$, then W' is linked over its own projections. That follows from tools already present in §5 — pp-propagation, intersection, and factoring — together with one elementary fact the paper never states: $0_{\mathbf{C}}$ is not an absorbing subuniverse of $\mathbf{C}^2$ when $|C| \geq 2$, because absorption at position i forces the term not to depend on argument i, at every i, hence to be constant, and idempotence then forces $|C| = 1$.

The subrelation form also discharges, for free, the subdirectness hypothesis the box form needed. Proof paper: `lem:no-absorbing-diagonal`, `lem:absorb-linked-sub`, `haz:box-vs-sub`.

Contrast. The analogous step in `LEMPCBridgesAreTrivial` Case 2 *is* justified by the source — by first proving the closure property $\xi \cap (A^2 \times B \times A) \subseteq B^4$ (`SS:1548-1554`), which makes $\mathrm{pr}_{1,2,3}(\xi \cap B^4) = \mathrm{pr}_{1,2,3}(\xi) \cap (A^2 \times B)$. Here B is a strong subuniverse rather than a linkedness block, no closure property is available, and absorption has to do the work instead.

2.9 D15 — read δ for σ, and restore stability

`SS:2208-2224`. A congruence $\delta \supseteq \sigma$ is chosen maximal with $|B/\delta| > 1$, and B/δ is shown BA and centre free. To show δ irreducible the proof supposes $\delta = S_1 \cap \cdots \cap S_k$ with $S_i \supsetneq \delta$, finds an i with $B^2 \not\subseteq S_i$, and then:

> "By Lemma `LEMLeftLinkedStayFull` LeftLinked$(S_i \cap (B \times B))/\sigma = (B/\sigma)^2$. Hence $(S_i \cap (B \times B))/\sigma$ is a linked relation, which by Lemma `LEMLinkedImpliesBACenter` implies the existence of a BA or central subuniverse on B/δ."

Two things are wrong, and they are the same thing twice. The relation is factored by σ while the conclusion is about B/δ; those do not match, since $(S_i \cap B^2)/\sigma$ lives on B/σ. And properness is not established: `LEMLinkedImpliesBACenter` needs a *proper* subdirect relation, $B^2 \not\subseteq S_i$ gives only $S_i \cap B^2 \subsetneq B^2$, and factoring can close that gap.

Repair. Read δ for σ throughout the last four lines, which is plainly what was meant — B/δ was shown BA and centre free two lines earlier for exactly this use, and the stated conclusion already says B/δ. Properness then follows, but only from a clause the proof drops when it introduces the S_i: the definition of irreducible (`main:1237`) requires the S_i to be **stable under** δ. Stability makes S_i a union of products of δ-blocks, so $(S_i \cap B^2)/\delta = (B/\delta)^2$ would force $B^2 \subseteq S_i$. That clause is the only part of the definition of irreducibility doing work here. Proof paper: `lem:main-existence`, `haz:main-existence`.

This propagates: `LEMUbiquity` rests on this lemma, so D15 reaches one of the headline properties.

2.10 D16 — the conclusion is the whole lemma, and is asserted

`SS:2743`: σ linear on $\mathbf{A} \in \mathcal{V}_n$ with $\sigma^* = A^2$ implies $\mathbf{A}/\sigma \cong \mathbf{Z}_p$. The proof produces the perfect-linear witness $\zeta \leq \mathbf{A} \times \mathbf{A} \times \mathbf{Z}_p$ with $\mathrm{pr}_{1,2}(\zeta) = A^2$, sets $\xi(x,z) = \zeta(x,a,z)$, and then says in full: "Then ξ is a bijective relation giving an isomorphism $\mathbf{A}/\sigma \cong \mathbf{Z}_p$." That is the entire content of the lemma. What ζ gives directly is only $\xi^{-1}(0) = [a]_\sigma$.

Repair, by a different route. The lemma follows from a statement already repaired here. A bridge δ from σ to σ with $\widetilde{\delta} \supsetneq \sigma$ exists by the linear criterion; replace it by $\delta * \delta^{-1}$, symmetric and pair-reflexive (§2.7). Then `LEMNontrivialReflexiveBridgeImplies`(3) applies, and $\sigma^* = A^2$ is a congruence with the single block A, so $\mathbf{A}/\sigma \cong \mathbb{Z}_p^m$ as a group. Now $m = 0$ gives $\sigma = A^2$, impossible because an irreducible congruence is proper (§4.4); and $m \geq 2$ contradicts irreducibility, because $\mathbf{A}/\sigma$ is affine, so the coordinate congruences $\theta_i = \{(x,y) : x_i = y_i\}$ are linear subspaces of $(\mathbb{Z}_p^m)^2$, hence subuniverses of $(\mathbf{A}/\sigma)^2$, each properly above $0_{\mathbf{A}/\sigma}$ with $\bigcap_i \theta_i = 0_{\mathbf{A}/\sigma}$. Hence $m = 1$. That the group isomorphism is an isomorphism *of algebras* is the proposition of §4.1, and without it the statement does not typecheck against $\mathbf{Z}_p \in \mathcal{V}_n$.

This route uses only in-paper material, where the source's route imports [13, Cor. 8.17.1] — so the repair is **one fewer external dependency** than the source. Proof paper: `lem:linear-on-top`, `haz:on-top`.

Entry 2.2 (Salvage: the source's ζ really is a function)**.** Worth keeping, because it repairs half of the original argument and is needed if the perfect-linear machinery is used elsewhere. Put $t(x,y,z) = w(x,y,\ldots,y,z)$ with $n-2$ copies of y; on $\mathbb{Z}_p$ this is $x-y+z$, because $p \mid n-1$. Applying t to three tuples of ζ over the same (a_1,a_2) gives, by idempotence, that the fibre is closed under $x-y+z$, hence is a coset of a subgroup of $\mathbb{Z}_p$: empty, a singleton, or all of $\mathbb{Z}_p$. It is not all of $\mathbb{Z}_p$, since that fibre would contain 0 and a nonzero element, making (a_1,a_2) both σ-related and not; and it is nonempty on σ^* since $\mathrm{pr}_{1,2}(\zeta) = \sigma^*$. So ζ is the graph of a homomorphism $G\colon \mathbf{A}^2 \to \mathbf{Z}_p$ with $G^{-1}(0) = \sigma$.

2.11 D17 — maximal multi-type extension. Open

`SS:3113`. σ is maximal with $(C_1 \circ \sigma) \cap B_2 = \varnothing$; Case 1 supposes $E <_{\mathrm{s}} B_1/\sigma$ and argues:

> "By Theorem `THMMainStableIntersection` $E \cap B_2/\sigma \neq \varnothing$ and $E \cap C_1/\sigma \neq \varnothing$, hence $C_1/\sigma \cap B_2/\sigma \neq \varnothing$, which contradicts our assumptions."

The two nonemptinesses are fine — though they come from the intersection property applied to the BA-and-central $D \subseteq E$, not from the theorem cited. **But "hence" does not follow**: two sets each meeting E need not meet each other. The maximality of σ, which would be the natural extra ingredient, is not used anywhere in Case 1.

Why the source's Case 1 cannot be patched in place. The configuration *reproduces itself* on D: one has $D \lhd\lhd\lhd \bar{A}$, $\bar{C}_1 \cap D \lhd\lhd\lhd D$, $\bar{B}_2 \cap D \lhd\lhd\lhd D$, still disjoint, with $D \cap \bar{B}_2 \neq \varnothing$. Every consequence of the two facts survives the descent, so no iteration of them closes the case. The descent is not useless — $C_1 \cap D \leq^{A}_{\mathcal{M}T} D$ properly, so the hypotheses are inherited and an induction on $|B_1|$ is available — but it shrinks the ambient, while the conclusion's congruence family is indexed by the ambient ($\omega^* \supseteq B_1^2$) and the single call site (`main:2685`) consumes the original B_1 in one of its two branches.

The published repair, and why it fails. An earlier draft of this document passed to a *minimal* subfamily $F \subseteq \{\bar{C}_1^1, \ldots, \bar{C}_1^t\}$ with $\bigcap F \cap \bar{B}_2 = \varnothing$, "which exists because the whole family works", and then used the stable-intersection theorem — whose leave-one-out hypothesis is exactly minimality, and whose four conclusions have no room for S — to show no member of F is S-typed, so that the case evaporates rather than being contradicted.

The first line is wrong, and an external review found it. What the hypothesis gives is $(\bigcap_j C_1^j)/\sigma \cap \bar{B}_2 = \varnothing$, whereas $(\bigcap_j C_1^j)/\sigma \subseteq \bigcap_j (C_1^j/\sigma)$ with the inclusion generally strict. So $\bigcap F$ can acquire σ-classes that meet $\bar{B}_2$, and the subfamily need not exist. This is precisely the image-versus-intersection failure that §4.8 is about — catalogued in this document and then walked into. The lesson is recorded rather than tidied away.

Where it stands. *Two things have changed since the repair was withdrawn.*

First, the image-versus-intersection failure is now known *not* to occur among the type-T factors. If $\bar{C}_1^j <_T \bar{B}_1$ then the "moreover" clause of `LEMFactorByDelta` gives $\sigma \cap B_1^2 \subseteq \sigma_j \cap B_1^2$, and a two-line argument then shows $\bigcap_{j\in F} \bar{C}_1^j = (\bigcap_{j\in F} C_1^j)/\sigma$ for F the set of such j. So the trap that killed the published repair is confined to the $<_{\mathrm{S}}$ and $= \bar{B}_1$ factors.

Second, given that, everything else in the argument runs: the proof paper proves `lem:maximal-mult` in full from the single extra hypothesis $(P \circ \sigma) \cap B_2 = \varnothing$, where $P = \bigcap_{j\in F} C_1^j$. No minimal subfamily, no S-freeness split, and the ambient stays $\bar{B}_1$, which is what the conclusion needs.

So the whole of D17 is now one sentence: *the type-T factors alone already cut B_2 out.* That is strictly stronger than what the other hypotheses give, since $C_1 \subseteq P$.

The saturation identity $\delta \cap B^2 \subseteq \sigma_j \cap B^2$ that makes images commute with intersection is the "moreover" clause of the factorisation lemma, and it is available for a factor precisely when that factor lands in the type-T alternative — not when it lands in $<_{\mathrm{S}}$. So the S factors are the obstruction both to clause (m) of propagation and to the minimal-subfamily argument, and the argument is circular as written.

One fragment survives, and it points at the target. A factor with $\bar{C}_1^j <_{\mathrm{S}} \bar{B}_1$ contains a BA-and-central $\bar{D}$ of $\bar{B}_1$, which meets $\bar{B}_1 \cap \bar{B}_2$ by the strong-nonemptiness lemma; so $\bar{C}_1^j \cap \bar{B}_2 \neq \varnothing$ — *the* S *factors never help cut* $\bar{B}_2$ *out.* What is missing is that the type-T factors alone already cut it out. If that is proved, saturation applies to them, the intersection of their images is the image of their intersection, and the rest of the argument runs with the ambient still $\bar{B}_1$.

Proof paper: `lem:maximal-mult`, marked *open* in its status table. Nothing downstream of that lemma should be written until this is settled.

2.12 C10 — a citation that drops a case

`SS:208` states:

> "Suppose $R \leq_{\mathrm{sd}} \mathbf{A}\times\mathbf{B}$, $C = \{c \in A \mid \forall b \in B : (c, b) \in R\}$. Then one of the following holds: (1) C is a central subuniverse of $\mathbf{A}$; (2) $\mathbf{B}$ has a nontrivial binary absorbing subuniverse."

[15, Thm. 6.15] (p. 38) is verbatim this, except for a third item: "3. **B** has a nontrivial *projective* subuniverse." The word "projective" does not occur anywhere in the 2024 source — `grep -i projective` over all four of its TEX files returns nothing — so the notion was removed from the paper, not merely from this citation.

The two-case statement is true, and the elimination is an argument Zhuk himself runs elsewhere without ever connecting it to Theorem 6.15. The standing hypothesis (`main:1123`, restated at `main:1641` precisely for the statements §5 proves) makes **B** Taylor. By [15, Lem. 3.4], a nontrivial projective subuniverse that is not binary absorbing produces an essentially unary $\mathbf{U} \in \mathrm{HS}(\mathbf{B})$ of size ≥ 2 — which a WNU forbids. Zhuk runs this same move on this same trichotomy in the proof of his Theorem 4.15, step (4)⇒(5): "In case (5) Lemma 3.4 gives us a nontrivial binary absorbing subuniverse, which is a strong subuniverse, or an essentially unary algebra $U \in HS(A)$, which contradicts condition (4)."

The elimination should be stated as a lemma with a proof rather than smuggled into a citation, and the proof paper does that, with a direct two-line argument for "a WNU forbids an essentially unary quotient" in place of the source's route through WNU-blockers.

But the two-case form is true only because ∅ counts as central. Under our reading (§4.3) it is false: $\mathbf{A} = \mathbf{B} = \mathbf{Z}_p$ with R the graph of $x \mapsto x + 1$ is subdirect, has $C = \varnothing$, and $\mathbf{Z}_p$ has no nontrivial binary absorbing subuniverse at all. So a third alternative $C = \varnothing$ must be added. It is harmless: each of the nine call sites exhibits an element of C first.

Proof paper: `lem:no-unary-quotient`, `lem:central-relation`, `haz:central-relation`.

Entry 2.3 (Two obligations at each of the eight contrapositive call sites)**.** The call sites are `SS:268, 551, 606, 843, 928, 1008, 2024, 2079`. Eight run the lemma contrapositively against a *BA and center free* algebra, defined at `main:1363` as having "no proper nonempty binary absorbing subuniverse or proper nonempty central subuniverse". Two obligations are therefore live at each and discharged at none explicitly.

Properness of the centre. Nonempty central is not enough; $C \neq A$ must be shown. It is available at every site, and at three of them the witness is printed two lines earlier — the non-containments at `SS:533, 824, 903` — but never connected to the appeal. Sites `606, 1008, 2079` instead *use* the improper case, concluding that R is full; there the split is explicit.

Power to base. Alternative (2) yields a binary absorbing subuniverse of a *power*; the sites assert one on the base. That is `LEMBACenterSOnPowerImplies`, cited at `SS:2024` and omitted at `551, 843, 928`.

2.13 F — a full fibre from a surjective projection

Subsubcases 1B3 (`SS:545`), 2A3 (`SS:837`) and 2B3 (`SS:921`) each contain, verbatim modulo names:

> "Since $\mathrm{pr}_{1,2,\ldots,|A|}(R') = (B/\delta)^{|A|}$, there exists $d \in B_{m-1}/\sigma_m$ such that $(B/\delta)^{|A|} \times \{d\} \subseteq R'$."

Surjectivity of a projection does not produce a full fibre, and as stated this is a non sequitur.

It is nevertheless true, for a reason that explains the arity. Each such R' is cut out by a conjunction of unary and binary conditions on its first $|A|$ entries together with a condition relating each entry to the last coordinate — at `SS:837`, $\forall i, j : (a_i, a_j) \in \sigma \cap \omega$, $\forall i : a_i \in B_{m-1}$, and $(a_i, b) \in \sigma_m$ — and any such condition survives substituting entries for one another. And $|B/\delta| \le |A|$ because $B \subseteq A$. So the tuple enumerating B/δ with repetition already lies in $(B/\delta)^{|A|}$, its witness d serves, and every other tuple is a coordinate-substitution instance of it.

The exponent $|A|$ is load-bearing: it must be at least the number of blocks, so that one tuple can enumerate them. A rendering that normalises the arity away breaks all three proofs. Proof paper: `lem:full-fibre`, `haz:full-fibre`.

2.14 D18 — **the definition of a linear congruence is ill-typed**

`main:1373`, clause (3) of the definition of a linear congruence, reads verbatim:

> "there exist prime p and $S \le (\sigma^*)^{[4]}$ such that for any block B of σ^* there exists $n \ge 0$ with $(B/\sigma; S \cap (B/\sigma)^4) \cong (\mathbb{Z}_p^n; x_1 - x_2 = x_3 - x_4)$."

S is a relation on *elements* of A — it is a subalgebra of $(\sigma^*)^{[4]} \subseteq A^4$ — while $(B/\sigma)^4$ is a set of 4-tuples of σ*-classes*. The two do not intersect: they are sets of different things, and $S \cap (B/\sigma)^4$ is empty on inspection, or meaningless, depending on how literally one reads it.

Repair. Cut down inside A and then take the image: $(S \cap B^4)/\sigma$. That is well formed for any S, and it is plainly what is meant, since the isomorphism is with a relation on $\mathbb{Z}_p^n \cong B/\sigma$.

The alternative repair — restrict the image relation S/σ to $(B/\sigma)^4$ — gives the same relation, because S is stable under σ in each variable and its tuples meet no other block. But that stability is a *consequence* of clause (3) (it is what makes S a bridge, the source's own following remark), so it is not available while clause (3) is being read, and a definition may not lean on its own consequences. Proof paper: `def:linear-congruence`, `haz:linear-typing`, and the propagation into `lem:linear-bridge`.

Note that the corresponding clause of `LEMNontrivialReflexiveBridgeImplies` is typed *correctly* in the source — it writes $(B/\sigma; (\delta \cap B^4)/\sigma)$ — so the slip is local to the definition.

3 Statements restated

Five statements of the source are restated in the proof paper, not because they are wrong but because our conventions make the printed form say something else — or because the source's own proof gives more than its statement.

Statement	**Why**	**Restatement**
`LEMCentralRelationImplies` (`SS:208`)	True in the source only because $\varnothing$ counts as central, and the projective alternative is dropped (§2.12)	"$C = \varnothing$, or C is central in $\mathbf{A}$, or $\mathbf{B}$ has a proper nonempty binary absorbing subuniverse"
`LEMLinkedImpliesBACenter` (`SS:222`)	"There exists a BA or central subuniverse" is vacuous: every algebra is a binary absorbing subuniverse of itself	Insert *proper nonempty*; that is how it is used at `SS:1282`
The $<_s$ clause (`main:1522`)	Vacuously true for $D = \varnothing$	Require the witness D nonempty
`LEMIntersectALL`(i)	Its proof (`SS:2356`) establishes $B \cap D \dot{\lhd}\dot{\lhd}\dot{\lhd}^A D$ and then weakens it; three later proofs use the stronger form while citing (i)	State (i) as $B \cap D \dot{\lhd}\dot{\lhd}\dot{\lhd}^A D$ and derive $B \cap D \dot{\lhd}\dot{\lhd}\dot{\lhd} A$ as a corollary
`CORReverseHomomorphism`	Its type list is $\{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{l}, \textsc{d}\}$ where the stable-intersection theorem uses $\{\textsc{ba}, \textsc{c}, \textsc{s}, \textsc{l}, \textsc{pc}\}$	Fix one convention; harmless, since D subsumes L and PC

4 The legislated readings

Ten places where [14] admits more than one reading. The choice is invisible in prose and forced in a formalization, and in six of the ten it is not really a choice: one reading makes some statement of the source true or some proof valid and the other does not, so the "convention" is a reconstruction. Two of the ten are not conventions at all — one is a theorem and one is a missing lemma — and only two are free.

§	Item	Status
4.1	the algebra $\mathbf{Z}_p$	**a theorem**, proved
4.2	σ^* is a tolerance	decided by internal evidence
4.3	the empty set	decided; and it bites three statements
4.4	the empty family is allowed	decided; the two formulations agree only under it
4.5	$\lhd\lhd\lhd$ carries its chain	decided; §5.5 is the sharpest instance
4.6	"dimension"	**free**
4.7	"linked instance"	**free**
4.8	quotients of subsets	decided; and see D17
4.9	sentences beginning "Notice that"	decided; promote to lemmas
2.12	the dropped projective alternative	**a missing lemma**, proved

4.1 The algebra $\mathbf{Z}_p$: not a convention, a theorem

`main:1115` declares *"In the paper every algebra $\mathbf{Z}_p$ belongs to $\mathcal{V}_n$ for a fixed n, hence the algebra $\mathbf{Z}_p$ is uniquely defined"*, with $w^{\mathbf{Z}_p} = x_1 + \cdots + x_n \bmod p$. That operation is idempotent only when $n \equiv 1 \pmod p$, so the declaration is a constraint on p, and the source never says where the constraint comes from.

It comes from *specialness.* If $\mathbf{U}$ is affine over $\mathbb{Z}_p$ with $|U| > 1$ and w is idempotent, weak near-unanimity and special of arity n, then w acts as $\sum_i c_i x_i + d$; idempotence gives $\sum_i c_i = 1$ and $d = 0$; the WNU identities force all c_i equal to a single c with $nc = 1$; and specialness compares the coefficient of y on the two sides of $w(x, \ldots, x, y) = w(x, \ldots, x, w(x, \ldots, x, y))$ to give $c^2 = c$, so $c \in \{0, 1\}$, and $c = 0$ contradicts $nc = 1$. Hence $c = 1$, $n \equiv 1 \pmod p$, and w acts as the sum.

Rendering. Keep the two roles of $\mathbf{Z}_p$ apart: the abelian *group* $(\mathbb{Z}_p; +, -)$, which supplies the relation $x_1 - x_2 = x_3 - x_4$ in the definition of a linear congruence and carries no arity constraint; and the *algebra* $\mathbf{Z}_p = (\mathbb{Z}_p; x_1 + \cdots + x_n) \in \mathcal{V}_n$, which does. Only the second occurs at `main:1300` ($\zeta \le \mathbf{A} \times \mathbf{A} \times \mathbf{Z}_p$, a subalgebra of a product, which needs a common signature) and at `main:3484`. Proof paper: `prop:p-divides`, `cor:Zp-in-Vn`, `haz:Zp`.

The proposition is not decoration: it is needed twice, and in both places its absence leaves a statement that does not typecheck — it is what makes the definition of a perfect linear congruence well formed, and what upgrades the group isomorphism of D16 to an isomorphism of algebras.

4.2 σ^* is a tolerance

σ^* is defined (`main:1246`) as the minimal $\delta \le \mathbf{A} \times \mathbf{A}$ with $\delta \supsetneq \sigma$ and δ stable under σ. Nothing in that makes it transitive.

Rendering. σ^* is a reflexive symmetric subalgebra of $\mathbf{A}^2$ and *not* a congruence. Three facts need one line each and none is in the source: the family of such δ is closed under intersection unless the intersection is σ, which is exactly what irreducibility forbids, so a least element exists; $(\sigma^*)^{-1}$ is another such relation, so symmetry follows by minimality — used silently at `SS:1538` ("switching x_1 and x_2"); and reflexivity is $\sigma \subseteq \sigma^*$.

The internal evidence. That σ^* is a congruence is item (2) of the definition of a linear congruence (`main:1371`) and conclusion (1) of `LEMNontrivialReflexiveBridgeImplies` —

i.e. it is a property of linear congruences, proved, not a property of irreducible ones. Typing it as a congruence would make item (2) vacuous and collapse the linear/PC distinction. Proof paper: `prop:cover`, `rem:cover-not-cong`, `haz:irreducible`.

4.3 The empty set

Read syntactically, $\varnothing$ is a subuniverse of every idempotent algebra, is vacuously absorbing ($t(\varnothing, \dots, A, \dots, \varnothing) \subseteq \varnothing$ holds because the left side is empty) and vacuously central (the clause quantifies over $a \in A \setminus C$ against $\mathrm{Sg}(\varnothing) = \varnothing$).

Rendering. Follow `main:1629` — *"we do not allow empty subuniverses"* — and make $\lhd\lhd\lhd$, $<_T$, $\leq_T$ relations between *nonempty* sets, with dotted variants for the empty case. Three statements must then be restated, because they are true in the source only under the vacuous reading; they are the first three rows of §3.

4.4 The empty family is allowed

Is $\sigma = A^2$ irreducible? The source gives two formulations of irreducibility (`main:1237`), and *they agree only under the empty-family reading*, which settles the question.

- Formulation 1: σ is not an intersection of *other* stable subalgebras of $\mathbf{A}^2$. The empty intersection is A^2, so $\sigma = A^2$ is reducible.
- Formulation 2: there are no $S_1, \dots, S_k \leq \mathbf{A}/\sigma \times \mathbf{A}/\sigma$ with $0_{\mathbf{A}/\sigma} = \bigcap_i S_i$ and $S_i \neq 0_{\mathbf{A}/\sigma}$. With $k = 0$ this says $0_{\mathbf{A}/\sigma} \neq (A/\sigma)^2$, i.e. $\sigma \neq A^2$.

Forbid the empty family and formulation 2 no longer excludes $\sigma = A^2$ while formulation 1 still does.

Rendering. Allow $k = 0$. Then an irreducible congruence is automatically proper, σ^* exists, and [13]'s word "proper" — which [14] drops — is recovered as a consequence rather than reinstated by hand. Proof paper: `def:irreducible`, `prop:cover`(a).

4.5 $\lhd\lhd\lhd$ carries its chain

$C \lhd\lhd\lhd^A B$ is defined (`main:1588`) as the *existence* of a chain $C = B_m <_{T_m} \dots <_{T_1} B_0 = B$; several §5 proofs then speak of *"the dividing congruences coming from $C \lhd\lhd\lhd^A B$"* (`main:1600`), which is a function of the chain, not of the pair.

The sharpest instance is the $\mathrm{D} \times \mathrm{D}$ case of `LEMTwoStableIntersection` (`SS:1840`). The step works only if the chain witnessing $C_2 \lhd\lhd\lhd \mathbf{A}$ is the chain witnessing $B_2 \lhd\lhd\lhd \mathbf{A}$ extended by the single step $C_2 <_{\mathrm{D}(\sigma_2)} B_2$ — that is what makes the dividing congruences of $C_2 \lhd\lhd\lhd \mathbf{A}$ equal to those of $B_2 \lhd\lhd\lhd \mathbf{A}$ together with σ_2, which is the whole content of the appeal. The same case also applies the clause to C_2 rather than to B_2; against B_2 it does not follow, since both alternatives are consistent with the hypotheses and the "moreover" clause then says nothing.

Rendering. $\lhd\lhd\lhd$ carries its witness: a list of pairs (subset, typed step with its congruence). Statements that mention "the congruences coming from $C \lhd\lhd\lhd B$" take the chain as an argument, extension of a chain by one step is an operation on the data, and the propositional $\lhd\lhd\lhd$ is its truncation, used only where no congruence is named. Proof paper: `def:lll`, `haz:lll-chain`.

4.6 "Dimension": free

$\prod_i \mathbb{Z}_{q_i}$ with distinct primes is not a vector space over anything, so it has no dimension. Use the composition length of the finite abelian group, which for $\mathbb{Z}_p^n$ is n and agrees with the source wherever the source is meaningful, and prove additivity along short exact sequences. Nothing in §5 uses the notion; it is confined to the codimension-one argument, where the correct statement is that such a group splits as a product over primes of $\mathbb{F}_q$-vector spaces. Proof paper: `haz:step2-linear-algebra`.

4.7 "Linked instance": free

Two inequivalent definitions coexist: global connectivity of the value graph, in the informal claim, and the per-variable condition in §3.1. The second does not imply the first, because a fragmented instance is per-variable linked in each component.

Rendering. Use the §3.1 definition and carry non-fragmentation as a separate hypothesis wherever the informal reading was doing that work. This is not cosmetic: the retracted recursion-depth claim (§6.2) turned on exactly this pair of conditions at the SolveNonlinked call site. Proof paper: `def:instance-linked`, `haz:linked`.

4.8 Quotients of subsets

B/σ denotes the *image* $\{b/\sigma : b \in B\}$ in $\mathbf{A}/\sigma$, for B any subset of A and σ a congruence on $\mathbf{A}$ — not the quotient of B by a congruence of $\mathbf{B}$. Consequently $(C_1 \cap \cdots \cap C_t)/\delta \subseteq \bigcap_i C_i/\delta$ always, with equality *false* in general.

Rendering. Define the image operation once, prove only the inclusion, and flag every site that needs the reverse. The reverse inclusion is established on the fly inside clause (fm) of Propagation — correctly, from the saturation identity $\delta \cap B^2 \subseteq \sigma_i \cap B^2$ — and must be extracted as a standalone lemma with its hypothesis visible. Proof paper: `haz:quotient-rel`, and the (fm) clause of `lem:propagation`.

This is the item the invalid repair of D17 walked into (§2.11). It is worth restating what that costs: cataloguing a trap is not the same as being immune to it, and the only defence that worked was someone else reading the argument.

4.9 Sentences beginning "Notice that"

Four items, three of them from the source's own "Notice that" sentences, all used later as lemmas.

(a) `main:1378`: the relation S from item (3) of the definition of a linear congruence is a bridge from σ to σ with $\widetilde{S} = \mathrm{pr}_{1,2}(S) = \mathrm{pr}_{3,4}(S) = \sigma^*$. **Checked, true.** Clause (4) is $x_1 - x_2 = 0 \iff x_3 - x_4 = 0$ inside each block; clause (3) holds because some block has $m \geq 1$, else $\sigma^* = \sigma$; and $\widetilde{S} = \sigma^*$ because $x - x = y - y$ always. Consumed at `SS:1360`. Proof paper: `lem:linear-bridge`.

(b) `main:1386`: σ is irreducible, PC or linear iff $0_{\mathbf{A}/\sigma}$ is. Needed to justify the "factorize by σ and assume $\sigma = 0$" opening used in three §5.4 proofs. Proof paper: `lem:irred-quotient`.

(c) `main:1550`: the two formulations of s-freeness agree. Trivial on unfolding $<_{\mathrm{s}}$ (`main:1522`), but it must be written, since s-freeness is a hypothesis of clause (m) of propagation modulo a congruence. Proof paper: `lem:sfree-equiv`.

(d) Two assertions about rectangularity belong to the same list: that the parallelogram property implies rectangularity ("as it follows from the definitions"), and that the rectangular closure exists (asserted by the definite article in "*the* minimal rectangular relation"). Note also that "R is rectangular" unqualified — the form used in "rectangular closure" — is never defined in the source; it means that every variable is rectangular. Proof paper: `lem:rect-basics`.

4.10 What legislating cost

Six of the ten (§§4.2, 4.3, 4.4, 4.5, 4.8, 4.9) are decided by internal evidence. Two (§§4.6, 4.7) are free choices where the source is simply ambiguous and either reading can be made to work. Two (§4.1, §2.12) are not conventions at all and have been proved.

5 Steps performed silently

These are not defects. They are the places where the proof paper must write out an argument that [14] performs in a clause, and they are the bulk of the work: about two dozen of them against fifteen register entries. They are collected here so that a reader comparing the two documents can see where the extra text comes from and satisfy themselves that it is expansion rather than divergence.

5.1 Coverage of §5

§5 of [14] is 3 144 lines, 44 statements with proofs, 152 proof-dependency edges. All seven subsections have now been read.

Lines	Subsection	Proofs	Defects found
48–70	Additional definitions	0	—
71–346	Subuniverses of types BA, C, S	6	none
347–1021	Intersection property	6	none; F localised here
1022–1735	Properties of PC or linear congruences	9	D11, D12, D13, D14
1736–1927	Types interaction	2	none; one missing mirror case
1928–2296	Factorisation of strong subalgebras	5	D15
2297–3144	Proof of the remaining statements	16	D16, D17

The distribution is not an accident. §5.4 is the only part of §5 that manipulates bridges as objects in their own right rather than consuming them, and every one of its four defects is about which auxiliary property survives a construction — symmetrisation, restriction to a block, restriction to a strong subuniverse.

5.2 The four hedges

A mechanical sweep for hedging phrases over the thirteen printed pages of §5 returns four, an unusually low density: the same regex over other expositions in this subject fires many times per page. **All four are honest.**

Line	In	Disposition
`SS:146`	`LEMBACenterSImplyFactor`	"For $T =$ BA it is straightforward": it is one line — if $t(B, A) \subseteq B$ and $t(A, B) \subseteq B$ then the same t works on images. Only $T =$ C is genuinely imported.
`SS:177`	`LEMBACenterSOnPowerImplies`	"For $T =$ S just repeat the same proof *word to word*": no repetition is needed — see below.
`SS:1560`	before `LEMNoBridgeBetweenDifferentTypes`	"This case can be considered in the same way as Case 2": it is genuinely the mirror. $(a, b, a, b) \in \xi$ places b in the block and $\sigma^* \subseteq$ RightLinked places a and c; writing it out is mechanical.
`SS:2428`	`LEMFactorByDelta` region	"The inclusion $\subseteq$ is obvious": it is.

Entry 5.1 (The source undersells its own citation)**.** `LEMBACenterSOnPowerImplies` says $B <_T \mathbf{A}^n$ with $T \in \{\text{BA}, \text{C}, \text{S}\}$ implies some $C <_T \mathbf{A}$, and proves it by "For $T \in \{\text{BA}, \text{C}\}$ see [15, Lem. 6.24]. For $T =$ S just repeat the same proof word to word replacing BA by S." The cited lemma reads: *"Suppose R is a nontrivial strong subuniverse of $A_1 \times \cdots \times A_n$ of type $T \neq$ PC. Then there exists i such that A_i has a nontrivial subuniverse of type T."* It is already general in T and already for a product of distinct algebras — the power is a special case — and its proof is *type-uniform*: it takes $\mathrm{pr}_1(R)$ if that is proper and otherwise a fibre, and neither construction mentions the type, which enters only through the facts that projections and fibres preserve BA and preserve C respectively. So a single run of the proof produces one witness carrying *every* strong type R has at once; if R is simultaneously binary absorbing and central — which is what S means — the witness is too. The hedge disappears rather than being discharged three times. Proof paper: `lem:power-to-base`, `haz:power-to-base`.

5.3 §5.3, the intersection property

`LEMIntersectionPCLinearIsGood` (`SS:623`) is a 292-line proof with 20 distinct citations, by a wide margin the largest and most connected in §5. It was read in full and is **structurally sound**. Every appeal to the inductive hypothesis lands at strictly smaller $k+\ell$: part (s) uses (s) and (d) at $k+(\ell-1)$; subsubcase 2B2 uses (s) at $(n-1)+k$; 2B3 uses (d) at $(n-1)+k$; and $n \le \ell$ throughout. That is worth confirming, since a commented-out block shows an earlier draft inducted lexicographically on $(|A/(\sigma\cap\omega)|,\ k+\ell)$ and the live proof dropped the first component.

Seven items for the rendering. The proof paper now carries the expansion in full; `haz:intersection-draft` keeps the four of these that a reader of the finished proof still needs.

1. *Part (s): five properness splits, not one.* Throughout part (s) the text asserts *proper* containments where `LEMBACenterImplyIntersection` and `LEMBACenterSImplyFactor` give only $\subseteq$. None is a defect in the lemma, because every equality branch gives the conclusion outright, but each needs writing. Branch $\mathcal{T}_\ell = \textsc{d}$: if $(G\cap C_{\ell-1})/\omega_\ell = C_{\ell-1}/\omega_\ell$ then every ω_ℓ-block meeting $C_{\ell-1}$ meets $G\cap C_{\ell-1}$, in particular the block E with $C_\ell = C_{\ell-1}\cap E$, so $G\cap C_\ell \neq \varnothing$. Branches $\mathcal{T}_\ell \in \{\textsc{ba},\textsc{c}\}$ and $\mathcal{T}_\ell = \textsc{s}$ each carry two more, one per containment into $B\cap C_{\ell-1}$; there the equality branch makes one of the two sets contain the other and the conclusion is immediate.
2. *What Lemma 6.25 is doing in part (s).* Part (s) appeals to `LEMBACenterSPossible Intersections` to conclude nonemptiness, from a lemma whose conclusion is "$T_1 = T_2 \in \{\textsc{ba},\textsc{c}\}$", which by itself permits two disjoint binary absorbing subuniverses. The step is valid only because $<_{\textsc{ba},\textsc{c}}$ is a *conjunction* (`main.tex:1567`): one of the two sets carries both types, so the type opposite to the other set's is available and equality of types fails. Proof paper: `lem:ba-central-meets`, `haz:ba-central-meets`.
3. *Properness in 2A1 and 2B1* comes from the minimal choice of m (resp. n); nonemptiness from the constant tuples $(b/\delta,\dots,b/\delta)$ for $b\in B$, which lie in $S_{m,0}$ since $B\subseteq B_m$ and $(b,b)\in\sigma\cap\omega$.
4. *Subsubcase 2B3* writes the first alternative as "$= C_n/\omega_n$", which is stronger than the "size 1" the inductive hypothesis supplies. Correct, because $B\cap C\neq\varnothing$ and $C\subseteq C_n$ force $B_k\cap C_n\neq\varnothing$, so the single block is the one defining C_n; then $B_k\cap C_{n-1}\subseteq C_n$ and $S_{k,n-1}$, $S_{k,n}$ cut $(B/\delta)^{|A|}$ identically, contradicting minimality of n.
5. *Subsubcase 2A3 reads one alternative off a two-alternative lemma.* "`LEMSelf IntersectionPC` implies that $((B\circ\delta)\cap B_{m-1})/\sigma_m = B_{m-1}/\sigma_m$" (`SS:828`) is not an implication: that lemma offers B_{m-1}/σ_m *or* B_m/σ_m. The second is excluded, but by an argument the text does not give: $B_m = B_{m-1}\cap E$ lies inside one σ_m-class, so the second alternative makes the last coordinate of R' constant, and with $\mathrm{pr}_{1,\dots,|A|}(R') = (B/\delta)^{|A|}$ that yields $(B/\delta)^{|A|}\times B_m/\sigma_m \subseteq R'$ — which the minimal choice of m forbids. Sound, but a real step.
6. *The full-fibre step* at `SS:545, 837, 921` is register entry F (§2.13).
7. *Is minimality of the chains k,ℓ used? No.* **Settled, and the hypothesis is dropped.** Writing the proof out decides it in the negative and shows it could not have gone the other way: the inductive appeals in part (s), 2B2 and 2B3 are to *truncations* of the chain for C, and a truncation of a minimal chain need not be minimal, so a step consuming minimality would break the induction rather than use it.
8. *The corollary* writes $\delta = f^{-1}(\sigma)$ with f undefined; read f_1. Its conclusion admits "empty" where the lemma requires $B\cap C\neq\varnothing$, so the empty case needs its own line at every call site. Proof paper: `haz:cor-intersection`.

Three smaller lemmas in the same subsection, never opened before the final pass, are all sound. `LEMTotallySymmetricWithoutBACenter` (`SS:357`) compresses one step: "by the conditions of the lemma we have $(a,\dots,a,c),(a,\dots,a,a) \in R$" does not follow from one application of the shift, which deletes the *last* entry and so destroys c. What is really happening is that symmetry plus the shift lets one pass from a tuple with entry multiset M to any $M + \{u\} - \{v\}$ with $u, v \in M$; $k-2$ such steps delete the b_i and duplicate a. This deserves its own one-line lemma, because the same manoeuvre is needed again in `LEMTotallySymmetricRelationForIrreducible` — where Case 2's "it remains to apply the previous lemma to $R \cap (B/\sigma)^n$" needs $\mathrm{pr}_{1,2}(R \cap (B/\sigma)^n) = (B/\sigma)^2$, not merely $\mathrm{pr}_{1,2}(R) \supseteq (B/\sigma)^2$. Proof paper: `lem:multiset-shift`, `haz:shift`.

5.4 §5.4, the bridge lemmas

Beyond D11–D14, `LEMBlockOfGoodBridgeDoesNotHaveBAC` is sound in both cases, and uses five steps without stating them.

1. $\widetilde{\delta}$ is *reflexive*, because $0_A \subseteq \mathrm{pr}_{1,2}(\delta) \subseteq \widetilde{\delta}$, the first inclusion being clause (c). Everything reflexive downstream comes from this.
2. $\mathrm{pr}_{1,2}(\sigma \cap E^4) = \omega \cap E^2$ for a subuniverse E, which is what makes Case 1's choice of E the right one. It needs $\sigma(x_1, x_2, x_1, x_2)$ for $(x_1, x_2) \in \omega$ — pair-reflexivity again, from the corrected symmetrisation.
3. *The E of Case 1 exists.* Maximality of E is never used, so it suffices to take the block containing a and b of the transitive closure of a compatible reflexive symmetric relation on D whose components are those of the undirected ω-graph. Note that $(\omega \cup \omega^{-1}) \cap D^2$, which suggests itself, is *not* a subalgebra — a union of two subuniverses need not be closed; use $(\omega \circ \omega^{-1}) \cap D^2$, which is compatible, reflexive and symmetric, and has the same components because ω is reflexive.
4. The "without loss of generality" at `SS:1170` needs $\widetilde{\sigma}$ symmetric, which holds for $\widetilde{\delta} \circ \widetilde{\delta}^{-1}$.
5. $\{a\} \circ \omega$ is a subuniverse, because $\{a\}$ is one by idempotence, which is what makes it equal to A in Case 2.

The appeal to the relational stable-intersection corollary at `SS:1214` is **correct but not in the corollary's shape**. The three displayed boxes $\{a\} \times B \times C \times B$, $\{a\} \times C \times C \times C$, $\{a\} \times C \times B \times B$ are the tight boxes of three witnesses, whereas the corollary wants A in the freed coordinate. Widening each to $\{a\} \times A \times C \times B$, $\{a\} \times C \times C \times A$, $\{a\} \times C \times A \times B$ gives its hypothesis with $n = 3$ and all ambients A, legitimate because $\lhd\lhd\lhd$ is reflexive; and since all three types are C and $n = 3$, every alternative fails. The three witnesses are (a, b, a, b), (a, a, a, a), (a, a, b, b). Two smaller things: $C' \leq_{\mathrm{C}} C$ at `SS:1226` comes out of pp-propagation as $\leq_{\mathrm{C}} A$ and needs intersection to land in C; and the appeal there to $\mathrm{pr}_1(\xi) = A$ should be to $\{a\} \circ \omega = A$.

`LEMNontrivialReflexiveBridgeImplies` is otherwise sound: the uniqueness of σ^*, the upgrade from "every block of $\mathrm{LeftLinked}(\sigma^*)$ of size > 1 satisfies $B^2 \subseteq \sigma^*$" to $\sigma^* = \mathrm{LeftLinked}(\sigma^*)$, and the prime argument all check out. Note that the contradiction is with the *minimality of* σ^*, not with irreducibility directly; the source says "contradicts the irreducibility of σ". The distinction matters where σ^* is data attached to a proof of irreducibility. Proof paper: `haz:minimality-not-irred`.

`LEMPCCongruencePropertyInductiveStep` is sound outside D14, and was checked line by line. The $|A| = 1$ worry does not arise: clause (c) forces $\mathrm{pr}_{1,2}(\delta) \supsetneq 0_A$, hence $|A| \geq 2$. Case 2's two applications of the ternary absorbing operation are correct and were recomputed coordinatewise; the step needs *ternary* absorption where $T = \mathrm{C}$ supplies only centrality, so "central implies ternary absorbing" (`main:1350`) is an unstated citation. Case 3 needs a word for the equality case, since the absorbing-equality lemma hypothesises a proper containment while the conclusion is immediate when it is an equality.

`LEMPCBridgesAreTrivial` is sound, with two steps to write out, both in Case 1.

The counting step (`SS:1478`): "since $\mathrm{pr}_{1,2}(\delta) = \mathrm{pr}_{3,4}(\delta)$, for any $(a, b, c, d) \in \delta$ the elements c/σ and d/σ are also uniquely determined by a/σ and b/σ". What the case hypothesis gives is uniqueness in the *other* direction — it is a statement about $\delta * \delta^{-1}$ and the claim is about $\delta^{-1} * \delta$. The bridge between them is pigeonhole, and the cited equality of projections is what powers it: on σ^*/σ, which is finite, δ induces a bipartite relation whose left neighbourhoods are nonempty, pairwise disjoint (that is what the case says) and cover the right side; equal finite cardinalities force every neighbourhood to be a singleton.

The four combinations (`SS:1534`): the proof derives $\mathrm{pr}_{1,4}(\delta) = \sigma$ or $\mathrm{pr}_{1,3}(\delta) = \sigma$, and by the same argument with x_1, x_2 switched $\mathrm{pr}_{2,4}(\delta) = \sigma$ or $\mathrm{pr}_{2,3}(\delta) = \sigma$, then says "This completes this case". Two of the four combinations are the conclusion; the other two must be excluded, and they are, because either would give $\mathrm{pr}_{1,2}(\delta) \subseteq \sigma$ against $\mathrm{pr}_{1,2}(\delta) = \sigma^* \supsetneq \sigma$. The reverse inclusion in the conclusion comes from stability under σ. The "switching x_1 and x_2" appeal needs σ^* symmetric.

The auxiliary relations ζ_1, ζ_2 *are* defined in the source, at `SS:1486-1497`, and the proof paper reproduces those definitions rather than reconstructing them: ζ_1 carries eight existentially quantified variables and four conjuncts, and the argument is a *subcase split* on whether ζ_1 is a bridge — not a claim that both are. An external review was right that an earlier draft of the proof paper invoked them without definition, and an intermediate draft supplied two short formulas that were not the source's.

One gap at the Case 2 call: hypothesis (3) of the bridge definition for $\xi \cap B^4$ requires $\sigma^* \cap B^2 \supsetneq \sigma \cap B^2$, which is not argued. It is not needed — if it fails, clause (d) puts both pairs of every tuple in σ, and linkedness on a B that is not a σ-block produces $(x_1, x_1, x_3, x_3) \in \xi$ with $(x_1, x_3) \notin \sigma$ directly, which is the conclusion. But the lemma cannot be *cited* on a non-bridge; the case has to be inlined. Proof paper: inside `lem:pc-bridges-trivial`.

5.5 §5.5, types interaction

Both statements are correct. `LEMBridgeBetweenCongruences` omits one direction: clause (4) is an *equivalence*, and the proof establishes only $(x_1, x_2) \in \sigma_1 \Rightarrow (y_1, y_2) \in \sigma_2$. The converse is the mirror argument, and it is where the *other* half of the hypothesis $\omega \cap \sigma_1 = \omega \cap \sigma_2$ is spent. Also unstated: clause (3) needs $\sigma_1 \subseteq \mathrm{pr}_{1,2}(\delta)$ as well as the strictness witness, and $(a, b) \notin \sigma_2$ follows because $\omega \setminus \sigma_1 = \omega \setminus \sigma_2$.

`LEMTwoStableIntersection` is missing one case and six steps. The missing case is the mirror $T_1 = \textsc{d}$, $T_2 \in \{\textsc{ba}, \textsc{c}\}$: the hypotheses are symmetric so it is the same argument with the indices swapped, but nothing says so, where `SS:1805` does say "similarly" for the mirror of the $\textsc{s}$ case. The six steps, all short and all discharged by $C_1 \cap C_2 = \varnothing$:

1. $C_1 \cap B_2 <_{T_1} B_1 \cap B_2$ needs *properness*: if $C_1 \cap B_2 = B_1 \cap B_2$ then $B_1 \cap C_2 \subseteq C_1 \cap C_2 = \varnothing$.
2. `SS:1830`: factoring returns $\subseteq$ and can collapse a proper containment; properness holds because $(C_1 \cap B_2)/\sigma_2 = B_2/\sigma_2$ would make the σ_2-block defining C_2 meet C_1.
3. `SS:1817` writes the first alternative as $(B_1 \cap B_2)/\sigma_2 = C_2/\sigma_2$ where the clause supplies only size 1; which block it is follows from $B_1 \cap C_2 \neq \varnothing$.
4. The $\textsc{d} \times \textsc{d}$ case must apply the clause to C_2, not B_2 (§4.5).
5. The chain for $C_2 \lhd\lhd\lhd \mathbf{A}$ must be *chosen* to extend the chain for $B_2 \lhd\lhd\lhd \mathbf{A}$ (§4.5).
6. $(c_1, c_2) \in \omega \setminus \sigma_1$: membership in ω because each B_i lies in a single block of every dividing congruence of its own chain and $c_1, c_2 \in B_1 \cap B_2$; non-membership in σ_1 because $c_2 \in B_1$ and $c_2 \, \sigma_1 \, c_1 \in C_1$ would give $c_2 \in C_1 \cap C_2$.

Proof paper: `lem:bridge-between-congruences`, `lem:two-stable`.

5.6 §5.6 and §5.7

Beyond D15, D16 and D17, the remaining sixteen proofs are sound. The steps to supply:

- `LEMCenterCanBePushedIn` takes the "full" alternative of the intersection corollary without a word. "Empty" is excluded by $R\cap(B_1\times B_2)\neq\varnothing$. *"Size one" needs two lines*: the class would have to be $[c]_\delta$, since c has an R-partner in B_1; and then $B_2''=B_2'\cap E$ either has $E=[c]_\delta$, putting c in B_2'' against the minimal choice of B_2', or misses $[c]_\delta$ entirely, so $R\cap(B_1\times B_2'')=\varnothing$ against $B_2\subseteq B_2''$. Smaller: $S'=(B_1/\sigma)^{|A_1|}$ holds because a *single* good c witnesses every tuple, and $S''\neq\varnothing$ is where $R\cap(B_1\times B_2)\neq\varnothing$ is spent.
- `LEMLeftLinkedStayFull`: Case 2's restriction to $n=2^k$ is not cosmetic — $R_n=(R\circ R^{-1})^n$ is symmetric, so $R_{2n}=R_n\circ R_n^{-1}$, which is what makes LeftLinked of the auxiliary relation computable in one step. Unstated: the auxiliary relation is subdirect over $(B_1/\sigma)\times\mathrm{pr}_2$ of itself, not over $(B_1/\sigma)\times A_2$; and $R_n\supseteq 0_{A_1}$ by subdirectness.
- `LEMCongruenceEitherCutOrDoNothing` leans throughout on $B^2\subseteq\omega$ — every B_i in the chain lies inside a single block of its own dividing congruence — which is never stated and is used at least four times. Also unstated: $|B/\sigma|>1$, and that a path witnessing linkedness must contain a single step crossing σ-classes.
- `LEMFactorByDelta`: Case 2 is a chain of six steps of which the source states three, and the **"moreover" clause is never argued** — it holds because Case 2 gives $\delta\cap B^2\subseteq\delta'\cap B^2=\sigma\cap B^2$ and Case 1 never produces the type-T alternative. That clause is the saturation identity, and it is what §2.11 turns on.
- `LEMPropagation`: clause (ft) is discharged by citing the factorisation lemma, which is stated only for $T\in\{\textsc{pc},\textsc{l}\}$; the $\{\textsc{ba},\textsc{c},\textsc{s}\}$ cases need clause (fs) and a sentence. And clause (fm)'s $t=1$ base case is where the $\textsc{s}$-freeness hypothesis is *spent* — it kills the factorisation lemma's middle alternative, which a multi-type cannot absorb. Nothing says so.
- `CORPropagateToRelations`(e) needs two unstated steps: $C\circ\delta=B\circ\delta$ is excluded because $\delta\subseteq\sigma$ puts $C\circ\delta$ inside one σ-block while $|B/\sigma|>1$; and the $<_{\textsc{s}}$ alternative is excluded because the witness also lies in one σ-block.
- `LEMMultiplyByAllLinear` invokes the stable-intersection theorem for its Case 1/Case 2 dichotomy, and **no theorem is consumed**: walk down the chain and ask where the intersection becomes empty; it is empty at B_1 by hypothesis, so either it is already empty at A or there is a first step where it becomes so. Proof paper: `haz:multiply-citation`.
- `LEMPCOnTheTopIsEasy` rests on two things it does not state: that an idempotent algebra is polynomially complete iff every *reflexive* subalgebra of a power is a conjunction of equalities, and that $\mathbf{A}/\sigma$ is BA and centre free, without which the step $\{a_i\}<_{\textsc{pc}}\mathbf{A}/\sigma$ is not licensed. Proof paper: `haz:on-top`.
- Typographical: `LEMUbiquity`'s "block D of B/δ" should read "block D of δ"; `LEMCenterCanBePushedIn`'s $\neq$ at `SS:2014` should be $\not\subseteq$; `CORParallelogramPropertyForD`'s undefined C at `SS:1000` should be C_1; `LEMMaximalMultExtention`'s $\delta_i^*\supseteq B_1^2$ at `SS:3128` should be $\delta_i^*\supseteq(B_1/\sigma)^2$; and the instance-fragmented definition writes X_1 twice where the second should be X_2.

5.7 `LEMPreserveLinkdness`: a flag withdrawn

Flagged in an earlier survey on the grounds that the workhorse `LEMPreserveLinkdnessOneStepAUX` assumes both $R\cap(B_1\times C_2)\neq\varnothing$ and $R\cap(C_1\times B_2)\neq\varnothing$ while the target lemma assumes only $R\cap(B_1\times B_2)\neq\varnothing$. **Not a defect**: the flag conflates the auxiliary lemma with `LEMPreserveLinkdnessOneStep`, which is what the four-line proof actually invokes and which needs only $R\cap(B_1\times B_2)\neq\varnothing$ and $S\cap(C_1\times B_2)\neq\varnothing$.

The four-line proof is correct and unpacks to a two-phase argument the proof paper should write out. Decompose both multi-type reductions into chains. *Phase 1*: shrink coordinate 1 along its chain with B_2 fixed; at each step the needed $R \cap (B_1^{(j)} \times B_2) \neq \varnothing$ is the previous step's conclusion, and the needed $S \cap (B_1^{(j+1)} \times B_2) \neq \varnothing$ follows from $S \cap (C_1 \times C_2) \neq \varnothing$ by monotonicity. This yields $R \cap (C_1 \times B_2) \neq \varnothing$. *Phase 2*: the same on the transpose R^{-1}, whose rectangular closure is S^{-1}, shrinking coordinate 2 with C_1 fixed — legitimate because $C_1 \lhd\lhd\lhd A_1$. Proof paper: `lem:preserve-linked`.

6 Claims examined and withdrawn

Two claims were raised, survived a round of review, and turned out to be wrong. They are the most instructive entries in this document, because both failed in the same way and it is the way a defect list is prone to fail: reading the formal-looking part of a paper and not the sentence before it.

6.1 The claimed citation cycle in §5

The claim was a cycle Lemma 9 $\Rightarrow$ 7 $\Rightarrow$ 86 $\Rightarrow$ 85 $\Rightarrow$ Cor 22 $\Rightarrow$ Thm 21 $\Rightarrow$ Lemma 8 $\Rightarrow$ Lemma 9, closed at `SS:1214`, and that it "has to be broken before anything in §5.4 or §5.7 can be formalized".

Refuted mechanically. Every `\begin{proof}` in `main.tex`, `StrongSubalgebras.tex` and `necessaryClaims.tex` was parsed, attributed to its statement (including the `\newtheorem*` duplicates the paper uses to restate results proved later), and its `\ref`s extracted. **58 statements with proofs, 152 edges, no cycles.**

The named edge is real: `LEMBlockOfGoodBridgeDoesNotHaveBAC` does invoke the relational stable-intersection corollary at `SS:1214`, in its $T = \textsc{c}$ case. But it closes nothing. That corollary's transitive proof-closure is 25 statements and does not contain `LEMBlockOfGoodBridgeDoesNotHaveBAC`; its direct dependencies are Propagation and the stable-intersection theorem, and the latter's closure is 15 statements, also disjoint from it.

Consequence: §5 can be written in dependency order, and there is no hidden simultaneous induction to untangle. The claim would not have survived ten minutes of parsing.

6.2 The recursion-depth bound

The claim was that the depth bound $|A| + |\Gamma|$ of [13, Lem. 5.2] was "not established for the CheckTuple $\to$ SolveNonlinked path", and that "without the repair the algorithm is not polynomial". **Withdrawn.**

Lemma 5.2 bounds the depth by arguing that every recursion site except CheckWeakerInstance "reduce[s] all domains of size greater than 1". At the SolveNonlinked site this needs the instance to be neither linked nor fragmented — otherwise the linked-component congruence can be trivial on some domain, that domain does not shrink, and the bound degrades from the constant $|A|$ to $O(|A| \cdot n)$, which is fatal, since the total work is $\mathrm{poly}(N)^{\mathrm{depth}}$. The source does state the hypothesis and does discharge it. It is simply not in one place:

- the precondition "an instance that is not linked and not fragmented" is in the prose introducing the function (`1704.01914:892`), and *not* in the `Input:` line of its pseudocode;
- it is discharged at the call site (`:635`) by "Θ_0 is not fragmented because it is crucial" — itself a one-line argument the source does not give: a fragmented unsatisfiable instance has an unsatisfiable part, and no constraint of the *other* part is then crucial;
- the step from "not fragmented" to "*every* domain splits" — propagate the partition along a path, which exists by non-fragmentedness and is value-preserving by 1-consistency — appears once, at `:1070`, inside the proof of a *different* lemma, and is never cited here.

So the mathematics is present and the theorem is not in doubt. What is absent is any single place where the six recursion sites of Lemma 5.2 are discharged. The same pattern holds for the Γ-order argument, whose case (3) needs the projections to be unchanged under weakening, which holds because the instance is 1-consistent when CheckWeakerInstance runs — true, and unstated.

Chasing it down did leave one genuine hole, and closing it needed an argument that is not in the source. CheckTuple calls SolveNonlinked not on Θ_0 but on Θ_0 with domains restricted, and restricting can only *destroy* the "not linked" precondition.

Lemma 6.1 (The SolveNonlinked call site)**.** *Let Θ_0 be cycle-consistent, non-fragmented and not linked, and let Θ_0' be Θ_0 with each domain restricted to a subset $D_x' \subseteq D_x$. Then every domain of Θ_0 of size greater than one is strictly shrunk by the time SolveNonlinked(Θ_0') recurses into Solve.*

Proof. Write $\tau_x = \mathrm{LinkedCon}(\Theta_0, x)$, a congruence by [13, Lem. 6.2]. First, τ_x is proper at *every* variable. Since Θ_0 is not linked there are at least two connected components of the graph on $\bigsqcup_z \{z\} \times D_z$; and every component meets every variable, because from any (x, a) non-fragmentedness supplies a path to y and 1-consistency extends a along it. So each D_y is partitioned into at least two nonempty τ_y-blocks, each therefore proper.

Now split on whether Θ_0' is linked.

Not linked. Then SolveNonlinked recurses on a component, whose trace on each D_y is a τ_y-block intersected with D_y' — a proper subset of D_y by the previous paragraph.

Linked. Then for every z and all $a, b \in D_z'$ some path of Θ_0' connects them; every such path is a path of Θ_0, so $D_z' \times D_z' \subseteq \tau_z$, i.e. D_z' lies inside a single τ_z-block. That block is proper, so $D_z' \subsetneq D_z$ — every domain has *already* been shrunk, at the CheckTuple step, before SolveNonlinked was called at all. □

Entry 6.2 (Why this is the right argument)**.** Two things are worth extracting.

First, the question this replaces was the wrong one. The natural worry is whether the restriction preserves the stated precondition "not linked", and one then starts asking how the minimal linear congruence ConLin relates to LinkedCon — a relation the source never states, and which is not needed. The dichotomy above uses no relation between them: *either* the precondition survives and the component split does the shrinking, *or* it fails and the restriction has already done it.

Second, it follows that SolveNonlinked's precondition is not required for the depth bound at all. The function is correct on a linked instance too — there is then one component and it recurses on the whole of it — and the lemma covers that case. The precondition should be read as documentation of the intended use, not as an obligation the caller must discharge.

6.3 Declared moot: the arity in the special-WNU lemma

[10, Lem. 4.7] is cited for the existence of a special idempotent WNU of arity $n^{n!}$ in $\mathrm{Clo}(w)$. The concern raised was that the period of $y \mapsto w(x, \ldots, x, y)$ need not divide $n!$. The suggested counterexample — $n = 3$ with a 5-cycle — does not work, since idempotence makes x a fixed point of that map and a full 5-cycle on five elements is then impossible; but cycles of length up to $|A| - 1$ remain possible, and $|A| - 1$ need not divide $n!$ when $|A| > n$.

Moot. The dichotomy proof consumes only "there exist N and a special idempotent WNU of arity N in $\mathrm{Clo}(w)$". The precise arity matters only for §4 of [14], which is an independent second result and out of scope. The proof paper states the weaker form.

6.4 A cycle in our own arrangement

The parser that refuted the claimed cycle in the source (§6.1) was pointed at the proof paper, and found one there:

`cor:intersection-good` → `lem:propagation` → `lem:factor-by-delta` →
`lem:cut-or-nothing` → `lem:leftlinked-stay-full` →
`lem:center-pushed-in` → `cor:intersection-good`

Every edge was a real citation. The cycle was *ours*: the source proves the corollary from `LEMReverseHomomorphism` (`SS:953`), a separate and much easier statement, where our draft had folded the backward clauses into Propagation and cited the whole thing. Citing the bundle drags in the forward clauses, which rest on §5.6, which rests back on the corollary.

The fix was to state the backward clauses as their own lemma, which the external review had independently asked for on different grounds — it noted that "the reverse-homomorphism lemma used for (bt) is neither stated nor cited". The proof paper now has `lem:reverse-hom`, and its graph is acyclic: 97 statements, 52 with proofs, 125 edges.

Two things worth drawing out. First, the source was right and we were wrong, which is the opposite of the direction this document usually runs. Second, this is the third time the same tool has settled a question in minutes that prose inspection had not settled at all.

Fixing the first cycle exposed a second, and this one the review had predicted. It wrote: "There is also a possible dependency-cycle danger: `block-good-bridge` invokes stable intersection; stable intersection invokes `no-cross-bridge`; and the most natural proof of no-cross appears to route through the linear bridge characterization and the block-good machinery. An explicit acyclic dependency graph is needed." That is exactly right, and once the two bridge-classification lemmas were written out the parser closed the loop:

`cor:stable-intersection` → `thm:stable-intersection` → `lem:bridge-to-pc` → `lem:no-cross-bridge` → `lem:linear-equiv` → `lem:nontrivial-bridge` → `lem:block-good-bridge` → `cor:stable-intersection`

It is not vicious. `LEMBlockOfGoodBridgeDoesNotHaveBAC` appeals to stable intersection at $n = 3$ with all types $\textsc{c}$, and that case rests on [15, Lem. 6.11] and on nothing in the bridge theory. The proof paper therefore states that case as its own lemma, proved before the bridges, and both consumers cite it.

The source avoids the loop by not citing: its `LEMPCBridgesAreTrivial` contradicts PC-ness without naming `LEMLinearEquivalentConditions`, so the edge that closes our cycle is absent from its graph. The dependency is nevertheless there in the mathematics — "σ is PC, so no bridge from σ to σ has $\widetilde{\delta} \supsetneq \sigma$" *is* that lemma — which is worth recording, because it means the earlier acyclicity verdict on the source (§6.1) is a statement about its citations and not about its logical dependencies. The claim refuted there was a specific named cycle, and that refutation stands; the stronger reading, that no cycle of any kind is latent in §5, was never established and is now known to be delicate.

7 The machine checks

Everything reducible to a finite check has had one. Each entry gives the scope, so that a reader can see what it does and does not cover. Two of the original claims of this document died to exactly such checks.

Check	**Scope**	**Result**
Proof-dependency graph, the source	58 statements, 152 edges over the three TeX files	Acyclic; §6.1
Proof-dependency graph, the proof paper	97 statements, 52 with proofs, 125 edges	Acyclic — after one real cycle was found and fixed, §6.4. Run by the build gate
Statement audit	all 81 labelled statements	9 flagged, 8 cleared by hand; D2 is the sole survivor
C1: special idempotent affine WNU on $\mathbb{Z}_p$	$p \in \{2, 3, 5, 7\}$, $3 \le n < 12$	exists exactly when $p \mid n-1$, and is then the sum
D6: the third hypothesis can fail	$\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_4$	witness found; §2.4
D12: counterexample	all six bijections $\mathbb{Z}_3 \to \mathbb{Z}_3$, exhaustive over $\mathbb{Z}_3^4$	none carries δ to the difference relation
D14: the subrelation form	nine small Taylor algebras (min-semilattices, the dual discriminator, $\mathbb{Z}_2$ minority, $\mathbb{Z}_3$, $\mathbb{Z}_2^2$, a semilattice square), $527 + 99\,477$ pairs (W, W')	no counterexample; and 0_C absorbs C^2 in no algebra tested

Entry 7.1 (What the searches are and are not)**.** A search over nine small algebras is evidence that a statement is not *obviously* false, and nothing more; the two positive entries above (C1 and D14) are backed by written proofs in the proof paper, and the searches were run first, to decide whether the proofs were worth attempting. The two *negative* entries are different in kind: a counterexample found by exhaustion over a finite structure is a proof, and D6 and D12 rest on nothing else. Those two are stated in the proof paper in a form a reader can check by hand in a few lines, which is the right standard for a paper; the exhaustive verification is a check on the transcription, not the argument.

Where the proof paper still says "verified by exhaustion" for a claim with no short symbolic proof, that is a gap, and it is listed in §9.

8 Disposition of the external review

An external reviewer read a single-file build of an earlier draft. Its findings, and what was done with each.

8.1 Accepted

- D17 **is broken at its first line.** Correct, and it reverses a claim made here. §2.11; the item is reopened.
- The symmetrisation lemma cited bridge composition, which carries an irreducibility hypothesis it does not have. Fixed by proving the collapse identity directly.
- The nice-bridge lemma asserted that $\delta * \delta$ is an equivalence relation, which needs a stabilised power or the Maltsev route.
- The block-good-bridge proof treated $(\omega \cup \omega^{-1}) \cap D^2$ as a subalgebra, which a union is not; use $(\omega \circ \omega^{-1}) \cap D^2$ (§5.4).
- "Which we may take reflexive" is used twice with no lemma behind it. Accepted as a gap in our exposition, but the gap is not where either of us thought: there is nothing to normalise. Every consumer hypothesises $\widetilde{\delta} \supsetneq \sigma$, which contains $\sigma \subseteq \widetilde{\delta}$, and $(a, a) \in \sigma \subseteq \widetilde{\delta}$ *is* $(a, a, a, a) \in \delta$. So the bridge is reflexive already, and the proof paper says so in one line (`lem:bridge-reflexive`). A genuine normalisation of the same *shape* does survive elsewhere — "we may assume $\widetilde{\delta}$ rectangular, as otherwise compose it with itself many times", in `LEMNoBridgeBetweenDifferentTypes` — and that one is open.
- The abelian-bridge characterisation needs $|A| > 1$, and "composing enough times" needs an explicit stabilised power.
- The reason given for reflexivity in `lem:perfect-from-linked` is false: (a, a) lying in the first projection gives some $\delta(a, a, c, d)$, not $\delta(a, a, a, a)$. The repair is the reviewer's: clause (d) gives $c\,\sigma\,d$ and stability replaces d by c.
- The document had the wrong shape, mixing a proof paper with a project status report. That is what this three-way split answers.
- A duplicate label, several overfull boxes, a `\ref` printed inside a `\cite`, and an unidentifiable bibliography entry. All fixed; the build gate now fails on each class.

Several further findings concern statements the proof paper had not yet written in full — the induction measure across coverings (Entry 2.1), weakening's termination order, the common binary absorption witness (§2.1), the mixed-prime linear algebra, and the definition of Solve. They are scheduled work rather than audit entries, and they are listed in the proof paper's status table.

8.2 Accepted, with a correction

The reviewer wrote that LeftLinked and RightLinked "are used throughout the strong-subalgebra proofs ... but no definition is given in the manuscript, and Brady's notes do not use those exact names". Both halves are true of *our* draft and of [2]. But [14] **does** define them, at `SS:52-66`, and in the general n-ary form:

> "For a relation $R \leq \mathbf{A}_1 \times \dots \mathbf{A}_n$ by LeftLinked(R) we denote the minimal equivalence relation on $\mathrm{pr}_1(R)$ such that $(a_1, a_2, \dots, a_n), (b_1, a_2, \dots, a_n) \in R$ implies $(a_1, b_1) \in$ LeftLinked(R). Similarly, RightLinked(R) is the minimal equivalence relation on $\mathrm{pr}_n(R)$ such that $(a_1, \dots, a_{n-1}, a_n), (a_1, \dots, a_{n-1}, b_n) \in R$ implies $(a_n, b_n) \in$ RightLinked(R)."

So this is not a defect of the source but an omission of ours: we reproduced the uses without the definition. It is now `def:linked` and `lem:linked-basics`, with the equivalence to connectedness of the bipartite graph proved for subdirect binary relations.

The n-ary form settles a question that matters elsewhere. LeftLinked splits R at its *first* coordinate and RightLinked at its *last*, with all intermediate coordinates going to the other side — so RightLinked($\mathrm{pr}_{1,2,3}(\delta)$) concerns the split $(1, 2) \mid (3)$ and is a relation on the third coordinate. That is exactly the reading D14 (§2.8) turns on, and it is now confirmed against the definition rather than inferred from the uses.

8.3 Rejected, with reasons

- **The witness in `LEMBridgeBetweenCongruences` is correct as printed.** The reviewer proposed reading $z_i = y_i = x_i$ for $z_i = y_i = x_1$. The printed choice works: it gives $(x_1, x_2, x_1, x_1) \in \delta$ using symmetry of σ_1 and reflexivity of ω, which is what is needed for $\sigma_1 \subseteq \mathrm{pr}_{1,2}(\delta)$. The proposed choice needs $(x_1, x_2) \in \omega$, which is not available.
- **$<_{\mathbf{ba,c}}$ is a conjunction, not a disjunction.** `main:1567` reads "we write $C <_{\mathrm{BA,C}} B$ meaning that $C <_{\mathrm{BA}} B$ and $C <_{\mathrm{C}} B$". The reviewer is right that the notation was never defined in our document, and it now is; but the reading is "both".

9 Risk register

What is least well tested, stated so that a later reader knows where to push.

Open mathematics. D17 (§2.11); the rectangularity normalisation in `LEMNoBridgeBetweenDifferentTypes`; the induction measure across expanded coverings (Entry 2.1). These gate the statements that depend on them and are marked *open* in the proof paper's status table. The reflexivisation, listed here in an earlier draft, turned out to need no argument at all.

Least-tested repairs. D7(ii)'s reweakening and D6's cruciality corollary each touch the CSP-instance machinery rather than the algebra, and each has been checked against only the call site that motivated it.

Least-tested positive verdicts. The self-intersection lemma was read only where the central-relation call sites forced it open. §5.3's three small lemmas were opened last, and one of them is what splits the main case of the largest proof in §5.

Unread. `main.tex`'s CSP half — instances, coverings, cruciality, the main induction, the algorithm — has been read only where D6, D7 and D8 touch it. That is roughly two thousand lines to which the standard of §1.1 has not been applied, and the register above should not be read as covering them. §4 of the source (`XYSymmetric.tex`, 1 640 lines) is out of scope by decision.

Claims resting on exhaustion alone. None of the register's positive claims, but the proof paper still carries "verified by exhaustion" where a short symbolic argument would serve — for instance the $\mathbb{Z}_3$ counterexample of D12, where setting $(x_1, x_2, x_3, x_4) = (0, b, b, 0)$ forces $2(f(0) - f(b)) = 0$, so f is constant for odd p and no bijection can work. Replace them.

Entry 9.1 (The failure mode this document is prone to)**.** Three times now a claim here has been published, reviewed, and turned out to be wrong: the citation cycle, the recursion-depth bound, and the repair of D17. In each case the error survived internal review and died to someone reading the argument from outside. A list like this one is exactly as reliable as the reading behind each entry, and the failure mode is not carelessness — it is confidence in a step that was never read at the standard of §1.1. The entries most at risk are the ones in the *unread* row above.

References

[1] A. Bulatov, P. Jeavons, A. Krokhin, *Classifying the complexity of constraints using finite algebras*, SIAM J. Comput. **34** (2005), 720–742.

[2] Z. Brady, *Notes on CSPs and polymorphisms*, arXiv:2210.07383. Cited by section title and numbered statement; the sections used here are “Cores and Idempotent Reducts” and §3.11, §3.15.

[3] L. Barto, M. Kozik, *Absorbing subalgebras, cyclic terms, and the constraint satisfaction problem*, Log. Methods Comput. Sci. **8** (2012), 1:07.

[4] L. Barto, A. Kazda, *Deciding absorption*, Internat. J. Algebra Comput. **26** (2016), 1033–1060.

[5] C. Bergman, *Universal algebra: fundamentals and selected topics*, CRC Press, 2011.

[6] A. Bulatov, *A dichotomy theorem for nonuniform CSPs*, arXiv:1703.03021; FOCS 2017.

[7] T. Feder, M. Y. Vardi, *The computational structure of monotone monadic SNP and constraint satisfaction*, SIAM J. Comput. **28** (1999), 57–104.

[8] D. Hobby, R. McKenzie, *The structure of finite algebras*, Contemp. Math. 76, Amer. Math. Soc., 1988.

[9] M. Istinger, H. K. Kaiser, *A characterization of polynomially complete algebras*, J. Algebra **56** (1979), 103–110.

[10] M. Maróti, R. McKenzie, *Existence theorems for weakly symmetric operations*, Algebra Universalis **59** (2008), 463–489.

[11] L. Barto, Z. Brady, A. Bulatov, M. Kozik, D. Zhuk, *Minimal Taylor algebras as a common framework for the three algebraic approaches to the CSP*, LICS 2021.

[12] R. Willard, *Similarity, critical relations, and Zhuk’s bridges*, slides, AAA98 — 98th Workshop on General Algebra, Dresden, 2019. `http://www.math.uwaterloo.ca/~rdwillar/documents/Slides/willard-aaa98-handout.pdf`

[13] D. Zhuk, *A proof of the CSP dichotomy conjecture*, J. ACM **67** (2020), no. 5, 1–78; arXiv:1704.01914.

[14] D. Zhuk, *A simplified proof of the CSP dichotomy conjecture and XY-symmetric operations*, arXiv:2404.01080v2.

[15] D. Zhuk, *Strong subalgebras and the constraint satisfaction problem*, J. Mult.-Valued Logic Soft Comput. **36** (2021); arXiv:2005.00593.